

Coarse Revenue Guarantee in First Price Auction and Beyond*

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September 4, 2026

Abstract

I analyze the coarse revenue guarantee (CRG) of the first-price auction with common values: CRG is the minimum expected revenue over Bayes Coarse Correlated Equilibria (BCCE), and BCCE is the equilibrium notion that no-regret learning leads to. The worst case is identical play, in which all bidders pool on a common bid, and this reduction yields a closed-form characterization. The guarantee is positive, increasing in the number of bidders, and strictly below the interim guarantee of [Bergemann, Brooks and Morris \(2017\)](#), converging to it at rate $\log N/N$. The analysis framework is extended to reserve prices, symmetric priors, other standard auctions beyond the first-price auction, and unknown value distributions.

*I am indebted to Songzi Du for his invaluable guidance on this project. I am immensely grateful to Aram Grigoryan, Jun Liu, Joel Sobel, and Emanuel Vespa for their useful suggestions for this paper. I also thank Cuimin Ba, Arjada Bardhi, Aislinn Bohren, Benjamin Brooks, Daniel Gottlieb, Duarte Gonçalves, Menglong Guan, Jiangtao Li, Markus Moller, Joao Ramos, Doron Ravid, Denis Shishkin, Vasiliki Skreta, Alex Smolin, Joel Watson, Frank Yang, and Mengxi Zhang, as well as the audiences at EC 24', GAIMSS 24', Stony Brook Game Theory Conference 2024, and UCSD TBE Workshop for valuable conversations and constructive comments. A previous version of this paper was circulated as "Revenue Guarantee across Bayes Coarse Correlated Equilibria." Claude is used for assistance with code optimization and text refinement. All errors are my own.

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1 Introduction

The robustness of auction mechanisms to the information structure is a central question in mechanism design. A seller who knows the distribution of the good’s value but not what the bidders know, or how the information is correlated, wants a lower bound on revenue that holds across every environment consistent with that prior. [Bergemann, Brooks and Morris \(2017\)](#) provide such a bound for the first-price auction. They characterize the **revenue guarantee** (RG), the minimum expected revenue across all information structures and all Bayes Correlated Equilibria (BCE), and show that it is positive, even as the number of buyers grows large.

The BCE framework retains one behavioral assumption. In a BCE, a bidder who is told to submit a_i after observing signal s_i must be unwilling to change that bid *at that signal*. Obedience holds state by state, which requires each bidder to process their private information and best-respond conditional on it. This is the natural requirement when bidders are sophisticated Bayesian agents.

It is not the appropriate requirement when bidders are algorithms. In modern electronic marketplaces, and increasingly in advertising exchanges and high-frequency trading venues, bids are submitted by automated agents. These agents adjust their bids over time by simple learning rules rather than by solving a Bayesian inference problem. The standard performance criterion for such rules is *no regret*: the algorithm guarantees that, in hindsight, no single fixed action played throughout the history would have earned more on average. That is an unconditional guarantee. It says nothing about whether the algorithm’s behavior would have been optimal in any particular state, and a no-regret algorithm can have large regret conditional on individual states without ever violating its guarantee.¹

The solution concept matched to this weaker notion of rationality is Bayes Coarse Correlated Equilibrium (BCCE), in which each player must be unwilling to switch to a single fixed action played in every state. [Hartline, Syrgkanis and Tardos \(2015\)](#) show that the empirical distributions generated by no-regret dynamics in a Bayesian game accumulate on the set of BCCE, not on the smaller set of BCE. A seller facing algorithmic bidders who wants a lower bound on revenue that holds for every outcome those algorithms can converge to must therefore minimize over BCCE. This paper

¹See [Lomys and Magnolfi \(2025\)](#) for an econometric framework built on exactly this weaker restriction, and [Nekipelov, Syrgkanis and Tardos \(2015\)](#) for inference from no-regret bidders.

carries out the analysis of [Bergemann, Brooks and Morris \(2017\)](#) for BCCE.

In this paper, I define the **coarse revenue guarantee** (CRG) as the minimum expected revenue across all BCCEs, and I ask four questions. How large is it in the first-price auction and when is it attained? How much does the seller lose by facing coarsely rational bidders rather than interim rational ones?² What can the seller do to raise it? And how do the results change across settings?

Characterization through identical play. The first contribution is a tractable characterization, and it is built on a structural result. In the first-price auction, from any BCCE one can construct another BCCE, with weakly lower revenue, in which every buyer submits the same bid with probability one, regardless of what they observe. The worst case is then an *identical play*: bidders do not use their private information and all submit the same bid.

The construction works because of two effects that act in the same direction. First, revenue in a first-price auction is the highest bid, so replacing a dispersed bid profile by a common bid lowers revenue. This effect holds under any notion of rationality. Second, the replacement can be arranged so that the distribution of the *highest competing bid* faced by any individual buyer is unchanged. This is why the construction preserves the equilibrium constraints. A coarse deviation is a single fixed bid, and the profitability of a fixed bid depends on one’s rivals only through the distribution of the best bid among them. Holding that distribution fixed while lowering the price means that each buyer’s payoff from deviating is unchanged while their payoff from obeying goes up. The constraints are thus relaxed, not violated.

This is different from the BCE benchmark. In [Bergemann, Brooks and Morris \(2017\)](#) the worst case is supported by an information structure in which bidders learn something about their rank, and play symmetrically *conditional* on that information. Under coarse rationality, the binding force is instead unconditional identical play. Bidders who cannot condition a deviation on their signal can be induced to submit a common bid that does not depend on the signal.

This reduction turns an N -dimensional problem into a one-dimensional one. In-

²The word *interim* refers to timing and is used throughout as the counterpart of *coarse*. In a Bayes correlated equilibrium a bidder decides whether to deviate after seeing their information, so the constraint is checked at the interim stage, conditional on the signal and the recommended bid. In a coarse equilibrium the deviation is one fixed bid chosen before any information arrives, so the constraint is ex ante, an average over all states. Section 3 gives the formal definitions.

stead of searching over joint distributions of values, signals, and N -dimensional bid profiles, the seller’s worst case is now determined by the joint distribution of the value v and a single common bid c . Theorem 1 solves this problem in closed form for an arbitrary value distribution. The guarantee is strictly positive for every non-degenerate prior and strictly increasing in the number of buyers.

The cost of coarseness. Because every BCE is a BCCE, the coarse guarantee is weakly below the interim guarantee of Bergemann, Brooks and Morris (2017), and Proposition 4 shows that the inequality is strict at every finite N , since no BCE ever attains the coarse minimum. The worst case has a rigid structure, and that structure cannot satisfy interim obedience. The gap is large when the number of bidders is small. With a uniform $[0, 1]$ prior and two buyers, the interim guarantee is $1/6 \approx 0.167$ while the coarse guarantee is 0.047, so the coarse guarantee is 72% below the interim guarantee. In the worst case, both buyers bid zero with probability 0.67, and the good is sold at a price of zero two-thirds of the time.

Theorem 2 shows that this gap closes as N grows. The coarse constraint is satisfied by lowering revenue until each buyer’s equilibrium payoff exceeds every fixed deviation’s payoff, and with N buyers sharing the surplus, that payoff is only $1/N$ of the total. As N grows, the revenue reduction needed to satisfy the coarse constraint goes to zero, and the two guarantees coincide in the limit.

Proposition 3 shows, however, that the convergence is slow. The interim guarantee approaches the common limit at rate $1/N$, the coarse guarantee only at rate $\log N/N$, so the cost of coarseness $RG_N - CRG_N$ itself vanishes at rate $\log N/N$. With a uniform prior and ten bidders, the coarse guarantee is still 19 percent below the interim guarantee. The asymptotic equivalence therefore holds only for large N .

What the seller could do. The next question is what the seller can do to increase the coarse guarantee. One instrument is a reserve price, under which the good is withheld if the highest bid is below the reserve. Section 6 shows that the identical play reduction holds with a reserve price, with a modified pooling construction, so the same tools apply, and the guarantee is characterized at every reserve level. Two findings follow. First, a small reserve raises the guarantee only slightly. One might expect any positive reserve to eliminate the states in which bidders receive the good at a price of zero. But the worst case now is pooling *at r* rather than at zero, so the good

still trades, the buyers pay r , and the surplus that supports their obedience falls only by r per state. The guarantee is continuous at $r = 0$ and rises at a rate proportional to the payments of r . Second, a larger reserve raises the guarantee substantially. As the reserve grows, pooling at the reserve becomes costly for the buyers, the worst case shifts from pooling at the reserve to withholding, and at a large enough reserve the pool at the reserve disappears. At that point the reserve eliminates the states in which the good is sold at a price of zero. In the two-bidder uniform example, the guarantee more than doubles at the revenue-maximizing reserve, from 0.047 to 0.106.

Beyond common values. The analysis is set in a pure common value model. Since the interim characterization of [Bergemann, Brooks and Morris \(2017\)](#) covers symmetric priors with arbitrary correlation, this setting is more restrictive, and [Section 7](#) examines what changes when values may differ across buyers. Within identical play, common values are the worst coupling: for any exchangeable prior with per-buyer marginal μ , every identical play BCCE yields revenue at least $CRG(\mu)$, because the rearrangement inequality behind [Theorem 1](#) uses only each buyer’s marginal, and the bound is attained exactly when the values coincide wherever bids are positive, because a single common bid can be aligned with every buyer’s value at once only if those values agree. But identical play is no longer without loss. When values differ, the adversary has an additional instrument. It can keep the price common and still allocate the good to the buyer with the highest value, by an arbitrarily small bid premium, and the resulting increase in the buyers’ total surplus relaxes the obedience constraints further. [Theorem 4](#) shows that the guarantee for any exchangeable prior is bounded below by the fixed point of [Theorem 1](#) with the mean replaced by the expected highest value, and that the bound is exact whenever values agree above the resulting threshold. Whenever values differ within the pooling region, the guarantee is strictly below the common value guarantee. Common values are therefore the worst coupling within identical play, but the worst symmetric priors overall agree at high values and differ at low ones.

Ranking and equivalence across auctions. [Section 8](#) compares the first-price auction with the other standard auctions, following [Bergemann, Brooks and Morris \(2019\)](#). The comparison rests on one observation. Under identical play every standard auction allocates by the same tie-break and charges the same price, so the outcome

attaining the first-price guarantee is available in every format, and formats differ only in what a single deviator pays after outbidding the others. When that payment is the deviator’s own bid, the identical play equilibria of the first-price auction are equilibria of the format, its guarantee restricted to identical play is the first-price guarantee, and its full guarantee is at most that. When the deviator pays the others’ bid instead, as in the second-price and English auctions, outbidding them does not raise the price paid, pooling at a low price cannot be sustained, and an asymmetric equilibrium yields a guarantee of zero. The first-price auction has the highest guarantee in both cases, so a seller designing for no-regret bidders is led to the same format as a seller designing for Bayesian bidders.

Unknown value distribution. Finally, Section 9 considers a seller who knows only the mean of the value distribution and wants a lower bound on revenue over BCCE. I first show that the coarse revenue guarantee is monotone with respect to the convex order: riskier priors, in the mean-preserving-spread sense, are worse for the seller. Because the binary distribution on $\{0, \bar{v}\}$ is the riskiest distribution on $[0, \bar{v}]$ with a given mean, it is the worst-case prior, and its guarantee is the lower bound. I then solve for this bound in closed form. The deviation constraint is satisfied because bids are correlated with values, and a mean-preserving spread weakens that correlation, which permits a lower revenue.

A caveat on interpretation. The worst case identified here requires two kinds of correlation, which differ in what they require. Bids must be correlated with one another across bidders, and bids must be correlated with the common value. The first requires little: the convergence of no-regret dynamics to the BCCE set concerns the empirical distribution of *joint* play, and a shared history of play produces correlation, since algorithms responding to the same past prices produce correlated bids without explicit coordination. The second requires more, because bids can be correlated with the value only if the bidders’ inputs carry value-relevant information, as they do on advertising exchanges where contextual features are correlated with value. An algorithm that uses such features and also has no regret *per context* converges to a smaller set than BCCE, and Remark 1 places the resulting guarantee between the two studied in this paper: the coarse guarantee is exact for algorithms with only unconditional no-regret guarantees, which is what standard no-regret algorithms

provide, and a valid lower bound for the rest. The design results hold either way. The reserve-price results of Section 6 improve the guarantee against *every* BCCE, correlated or not, so the bound holds for a seller who adopts them whether or not the worst case is reachable.

The remainder of the paper is organized as follows. Section 2 discusses related literature. Section 3 introduces the model and the two equilibrium notions. Section 4 establishes the identical play reduction. Section 5 characterizes the coarse revenue guarantee and establishes asymptotic convergence and its rate. Section 6 develops the reserve-price analysis. Section 7 extends the characterization to symmetric priors. Section 8 compares the guarantee across standard auctions. Section 9 treats an unknown value distribution. Section 10 concludes. Omitted proofs are in the Appendix.

2 Related Literature

This paper contributes to three strands of literature: first-price auctions, robust revenue guarantees, and the intersection of Bayes Coarse Correlated Equilibrium with no-regret learning dynamics.

First, this work builds on the literature on the first-price auction. Early studies, such as Vickrey (1961), Milgrom and Weber (1982), Battigalli and Siniscalchi (2003), Dekel and Wolinsky (2003), Fang and Morris (2006), and Azacis and Vida (2015), relied on specific assumptions about the information structure. More recently, Bergemann, Brooks and Morris (2017) allowed for arbitrary information structures but restricted attention to Bayes Correlated Equilibrium. This paper characterizes the coarse revenue guarantee under the weaker notion of BCCE, and by developing coarse counterparts of the ranking and equivalence results of Bergemann, Brooks and Morris (2019). The closest earlier work is Feldman, Lucier and Nisan (2016), who study correlated and coarse equilibria of single-item auctions. Their focus is on welfare in the full-information model, whereas I minimize revenue over all information structures and characterize the minimizing equilibrium exactly.

Second, this paper contributes to the literature on revenue guarantees, including Bergemann, Brooks and Morris (2017), Bergemann, Brooks and Morris (2019), Du (2018), and Brooks and Du (2021), by investigating these guarantees outside the scope of BCE. While there is existing work studying guarantees in terms of welfare,

specifically the price of anarchy (Syrkanis and Tardos, 2013; Jin and Lu, 2023), this paper focuses strictly on revenue rather than welfare outcomes. The reserve-price result of Section 6 also connects this paper to the literature comparing competition with mechanism design initiated by Bulow and Klemperer (1996), whose theorem states that an auction with one additional bidder dominates the optimal mechanism with a reserve price. Against coarsely rational bidders a small reserve raises the guarantee only slightly, because bidders pool at the reserve and pay it. A large gain requires a reserve large enough that the worst case involves withholding rather than pooling at the reserve.

Finally, this work contributes to the literature on BCCE and learning in games. BCCE was first proposed by Moulin and Vial (1978). It received little attention in economics because its set of deviations is small. The use of no-regret algorithms as bidders has renewed interest. As shown by Hartline, Syrgkanis and Tardos (2015) and supported by Hart and Mas-Colell (2000) and Blum et al. (2008), no-regret dynamics converge to the set of BCCE. A growing literature studies auctions in which some or all participants are learning algorithms: Nekipelov, Syrgkanis and Tardos (2015) on inference from no-regret bidders, Braverman et al. (2018) and Deng, Schneider and Sivan (2019) on a seller’s ability to exploit them, Kolumbus and Nisan (2022) on auctions between learning agents, and Lomys and Magnolfi (2025) on structural estimation under no-regret restrictions. Relative to that literature, which typically fixes a learning protocol or an information structure, I take the set-valued approach of robust mechanism design and bound revenue over the entire BCCE set.

3 Model

3.1 Setup

Consider a single-unit auction with a set of risk-neutral buyers $\mathcal{N} = \{1, \dots, N\}$, where $N \geq 2$. The seller owns one indivisible good and values it at zero. The buyers have a pure common value for the good, denoted $v \in V = [0, \bar{v}] \subseteq \mathbb{R}_+$, distributed according to a probability measure $\mu \in \Delta(V)$ with mean $m := \mathbb{E}_\mu[v]$ and cdf F . Section 7 treats general symmetric priors, for which the common value coupling is the worst case within identical play but not overall.

Assumption 1. μ admits a density f that is positive on $(0, \bar{v})$, and each buyer's action set is $A_i = [0, \bar{v}]$, so that $A = \prod_i A_i = [0, \bar{v}]^N$.

The density assumption is kept throughout to simplify the exposition, not because the results need it. Under it F is continuous and strictly increasing on $[0, \bar{v}]$, and F^{-1} denotes its inverse. Section 9 extends the characterization to arbitrary priors, including atomic ones, which is needed there because the worst-case prior among distributions with a given mean is binary.

Restricting bids to $[0, \bar{v}]$ is without loss of generality in the first-price auction, since a bid above $\bar{v}$ guarantees a loss whenever it wins. It is not without loss in Section 8, where the set of allowed deviations is exactly what distinguishes one format from another, so I state the restriction explicitly rather than leaving it implicit.

In the first-price auction (FPA), the highest bidder wins. If there are ties, the good is allocated with equal probability among the tying bidders. Formally, the allocation rule $q_i(a)$ for buyer i is

$$q_i(a) = \begin{cases} 1, & \text{if } a_i > \max_{j \neq i} a_j, \\ \frac{1}{|\{k \in \mathcal{N} : a_k = \max_j a_j\}|}, & \text{if } a_i = \max_j a_j, \\ 0, & \text{otherwise,} \end{cases}$$

and the winner pays their own bid. Buyer i 's ex-post utility given the common value v and the action profile a is

$$u_i(v, a) = q_i(a)(v - a_i).$$

Because the good is always allocated to somebody in the first-price auction, the buyers' payoffs and the seller's revenue always exhaust the value:

$$\sum_i u_i(v, a) = v - \max_i a_i \quad \text{for every } (v, a). \quad (1)$$

This accounting identity is used repeatedly below. Anything that lowers revenue raises the buyers' total payoff by exactly the same amount. Since the equilibrium constraints are constraints on buyers' payoffs, a construction that lowers revenue automatically makes obedience easier to satisfy, provided it does not simultaneously make deviations more attractive. Much of the analysis consists of constructing such

reductions.

3.2 Information and Two Notions of Obedience

The buyers' correlated information about the common value is described by an *information structure* $I = (S, \theta)$, where S_i is a finite set of signals for buyer i , $S = \prod_i S_i$, and $\theta \in \Delta(V \times S)$ is the joint distribution of values and signals. Let $\mathcal{I}(\mu)$ be the set of information structures whose value marginal is the prior, $\text{marg}_V \theta = \mu$. The seller does not know I , and the guarantee is computed over all $I \in \mathcal{I}(\mu)$.

Given I , a *decision rule* is a map $b : S \rightarrow \Delta(A)$. Equivalently, a mediator observes the signal profile s and privately recommends a bid a_i to each buyer i . A decision rule induces the joint distribution

$$\sigma_b(dv, ds, da) = b(da \mid s) \theta(dv, ds) \in \Delta(V \times S \times A),$$

and buyer i 's expected utility is $U_i(I, b) := \int u_i(v, a) \sigma_b(dv, ds, da)$.

The two solution concepts I compare differ in exactly one respect: *how much a buyer is allowed to know when choosing a deviation*. I state both.

Definition 1 (Bayes Correlated Equilibrium). *A decision rule b is a Bayes correlated equilibrium of I if for every buyer i and every measurable map $\xi_i : S_i \times A_i \rightarrow A_i$,*

$$\int u_i(v, a) \sigma_b(dv, ds, da) \geq \int u_i(v, \xi_i(s_i, a_i), a_{-i}) \sigma_b(dv, ds, da).$$

Write $\mathcal{BCE}(I)$ for the set of BCE of I .

Definition 1 is the standard obedience formulation of [Bergemann and Morris \(2016\)](#). A buyer contemplating a deviation may condition on two things: the signal s_i they observed, and the bid a_i they were told to submit, since the recommendation is informative about the mediator's information. Obedience must hold state by state. The map ξ_i may pick a different deviation in every contingency the buyer can distinguish.

Definition 2 (Bayes Coarse Correlated Equilibrium). *A decision rule b is a Bayes coarse correlated equilibrium of I if for every buyer i and every constant deviation $a'_i \in A_i$,*

$$\int u_i(v, a) \sigma_b(dv, ds, da) \geq \int u_i(v, a'_i, a_{-i}) \sigma_b(dv, ds, da).$$

Write $\mathcal{BCCE}(I)$ for the set of BCCE of I .

Here the buyer must commit, before seeing anything, to a single bid to be submitted in every state, and compares the *average* payoff from that commitment to the average payoff from following the mediator’s recommendations. A buyer in a BCCE may regret their bid in particular states, but no single bid yields a higher average payoff across all states. This is the guarantee that a no-regret learning algorithm provides, and it is why the empirical distributions generated by such algorithms accumulate on this set rather than on the smaller BCE set (Hartline, Syrgkanis and Tardos, 2015; Blum et al., 2008). In the language used throughout the paper, Definition 1 imposes *interim* obedience, evaluated after the buyer sees their information, while Definition 2 imposes *ex ante*, or coarse, obedience, evaluated before.

Because a constant map is one particular instance of a deviation map ξ_i , we have $\mathcal{BCE}(I) \subseteq \mathcal{BCCE}(I)$ for every I : BCCE is the more permissive concept. The following example illustrates the difference.

Example 1 (A BCCE that is not a BCE). Let $N = 2$ and let v take the values 0 and 1 with equal probability. Suppose both buyers observe v and the mediator recommends the common bid $c = v/3$, so that both bid 0 when $v = 0$ and both bid $1/3$ when $v = 1$. Each buyer’s expected payoff is $\frac{1}{2} \cdot \frac{1}{2} \cdot (1 - \frac{1}{3}) = 1/6$. A buyer deviating to the fixed bid $\delta \in (0, 1/3)$ wins only when $v = 0$, earning $\frac{1}{2}(0 - \delta) < 0$. A buyer moving to $\delta \in (1/3, 1)$ always wins and earns $\frac{1}{2}(0 - \delta) + \frac{1}{2}(1 - \delta) = \frac{1}{2} - \delta < \frac{1}{6}$. A buyer deviating to exactly $1/3$ wins when $v = 0$ and ties when $v = 1$, earning $\frac{1}{2}(0 - \frac{1}{3}) + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2}{3} = 0$. No fixed bid improves on $1/6$, so the profile is a BCCE, with the constraint binding at bids just above $1/3$. It is not a BCE: conditional on the recommendation $c = 1/3$, the buyer knows $v = 1$ and can raise the bid slightly to win, gaining nearly $\frac{1}{2} \cdot \frac{2}{3}$ instead of $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2}{3}$. The unconditional deviation is unprofitable only because the good states are averaged against the bad ones.

Example 1 illustrates the mechanism behind the paper’s results. Under coarse rationality, revenue can be lowered by equilibria in which bidders would gain from deviating in the states where the good is valuable, but are deterred by the losses they would incur in the states where it is not. The low bid is sustainable because the deviation payoff is averaged across states, and interim obedience does not average across states.

Remark 1 (Between coarse and interim obedience). The two definitions are the ex-

treme cases of a family, ordered by how finely a deviation may condition on information. Fix for each buyer a partition of their signal set and require obedience against deviations that are constant on each cell. Coarsening every partition to a single cell recovers Definition 2, refining to singletons yields signal-measurable deviations, and allowing deviations to condition on the recommendation as well recovers Definition 1. The equilibrium sets are nested and shrink as deviations refine, so worst-case revenue is monotone along the family, and the two guarantees studied in this paper bound the guarantee of every intermediate specification. This matters for the algorithmic interpretation: standard no-regret algorithms provide only the unconditional guarantee, while contextual algorithms, which take value-relevant features as inputs, provide per-context guarantees that lie strictly between the two. The coarse revenue guarantee is the exact lower bound for the former and a valid lower bound for the latter.

Let $R(I, b) := \int \max_i a_i \sigma_b(dv, ds, da)$ be expected revenue. The *revenue guarantee* and the *coarse revenue guarantee* under the prior μ are

$$RG(\mu) := \inf_{I \in \mathcal{I}(\mu)} \inf_{b \in \mathcal{BCE}(I)} R(I, b), \quad CRG(\mu) := \inf_{I \in \mathcal{I}(\mu)} \inf_{b \in \mathcal{BCCCE}(I)} R(I, b),$$

and since minimizing over a larger set yields a smaller value, $CRG(\mu) \leq RG(\mu)$ for every prior.

3.3 A Reduced Notation for the Coarse Problem

Definition 2 has a property that Definition 1 lacks. Because a coarse deviation cannot condition on the signal, the signals never appear in the constraint except through the joint law they induce over values and actions. Two information structures that generate the same joint distribution of (v, a) impose exactly the same coarse obedience constraints, even if one is more informative than the other. The information structure can therefore be dropped in favor of a joint distribution $\sigma \in \Delta(V \times A)$ whose value marginal is μ .

Lemma 1 (Reduced formulation). *A distribution $\sigma \in \Delta(V \times A)$ with $\text{marg}_V \sigma = \mu$ is the value-action marginal of some σ_b with $I \in \mathcal{I}(\mu)$ and $b \in \mathcal{BCCCE}(I)$ if and only*

if, for every buyer i and every $a'_i \in A_i$,

$$\int_{V \times A} u_i(v, a) d\sigma(v, a) \geq \int_{V \times A} u_i(v, a'_i, a_{-i}) d\sigma(v, a). \quad (2)$$

Proof. Necessity holds because both sides of Definition 2 depend on σ_b only through its $V \times A$ marginal. For sufficiency, let every buyer observe a null signal and let the mediator draw the bid profile from the conditional $\text{marg}_A \sigma(\cdot | v)$. The induced joint law is σ , and (2) is exactly the required obedience condition. $\square$

Write $\mathcal{BCC}\mathcal{E}(\mu)$ for the set of σ satisfying (2), and $R(\sigma) = \int \max_i a_i d\sigma$. Then

$$CRG(\mu) = \inf_{\sigma \in \mathcal{BCC}\mathcal{E}(\mu)} R(\sigma),$$

and any σ attaining this infimum is called a *min-R BCCE*.

I do not assume that such a σ exists. Because the payoff functions u_i are discontinuous at ties, the map $\sigma \mapsto \int u_i d\sigma$ is not continuous in the weak-* topology, and $\mathcal{BCC}\mathcal{E}(\mu)$ need not be weak-* closed, so the existence of a minimizer does not follow from compactness. Every argument below is therefore constructive. Each step maps a given BCCE to another BCCE with weakly lower revenue, and existence of the minimizer is established at the end by exhibiting it.

Ties require a separate argument, since the payoff functions are discontinuous at ties. The following lemma lets every subsequent argument ignore them. It says that for an identical play distribution it is enough to check the coarse constraint at bids which the buyers do not submit with positive probability, and these are dense.

Lemma 2 (Tie-free verification). *Let c be a random common bid submitted by all N buyers, and let K be each buyer's equilibrium payoff. Suppose that*

$$K \geq \mathbb{E}[(v - \delta) \mathbb{1}\{c < \delta\}]$$

for every δ in a dense subset $D \subseteq [0, \bar{v}]$. Then the coarse obedience constraint (2) holds at every $\delta \in [0, \bar{v}]$, ties included.

Proof. Fix δ and set $A := \mathbb{E}[(v - \delta) \mathbb{1}\{c < \delta\}]$ and $B := \mathbb{E}[(v - \delta) \mathbb{1}\{c \leq \delta\}]$. Deviating to δ wins when $c < \delta$ and shares with the $N - 1$ other buyers when $c = \delta$, so the

deviation payoff equals

$$A + \frac{1}{N}(B - A) = \left(1 - \frac{1}{N}\right)A + \frac{1}{N}B \leq \max\{A, B\}.$$

Choose $\delta_n \in D$ with $\delta_n \uparrow \delta$. Then $\mathbb{1}\{c < \delta_n\} \uparrow \mathbb{1}\{c < \delta\}$ and $\delta_n \rightarrow \delta$, so by dominated convergence $\mathbb{E}[(v - \delta_n)\mathbb{1}\{c < \delta_n\}] \rightarrow A$, giving $K \geq A$. At $\delta = 0$ this step is vacuous, and $K \geq A = 0$ follows instead from the downward limit below, since $B \geq 0$ there. Choose $\delta_n \in D$ with $\delta_n \downarrow \delta$. Then $\mathbb{1}\{c < \delta_n\} \downarrow \mathbb{1}\{c \leq \delta\}$, so the same expectations converge to B , giving $K \geq B$. At $\delta = \bar{v}$ this step is vacuous, and $K \geq B$ follows instead from signs, since $c \leq \bar{v}$ always gives $B = \mathbb{E}[v - \bar{v}] \leq 0$, while letting $\delta_n \downarrow 0$ in D gives $K \geq \mathbb{E}[v \mathbb{1}\{c = 0\}] \geq 0$. Hence $K \geq \max\{A, B\}$, which dominates the deviation payoff. $\square$

4 The Identical Play Reduction

Minimizing revenue directly over $\mathcal{BCC}\mathcal{E}(\mu)$, a set of distributions over N -dimensional bid profiles, is not tractable. This section shows that the minimization can be restricted to a one-dimensional family. Call σ an *identical play* distribution if it puts probability one on profiles in which all buyers submit the same bid, $a_1 = \dots = a_N$.

Proposition 1 (Identical Play Reduction). *For every $\sigma \in \mathcal{BCC}\mathcal{E}(\mu)$ there is an exchangeable identical play $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$ with $R(\gamma) \leq R(\sigma)$, and the inequality is strict unless σ already places probability one on profiles whose two highest bids coincide. Consequently*

$$CRG(\mu) = \inf \{R(\gamma) : \gamma \in \mathcal{BCC}\mathcal{E}(\mu), \gamma \text{ identical play}\}.$$

The formal argument is in Appendix A. Because the construction is used again in Section 6 in modified form, I describe it in words.

Why revenue falls. In a first-price auction the seller collects the highest bid. The maximum is a convex function of the bid profile, so for a fixed collection of bid levels the seller does best when bids are spread out and worst when they coincide. Pooling a dispersed profile onto a common level therefore lowers revenue. Revenue in a first-price auction comes from price competition among bidders, and bidders who submit

identical bids do not compete on price.

Why obedience survives. A construction that lowers revenue by reducing competition might also raise deviation payoffs. The pooling used here does not.

I use symmetrization first, so that all buyers can be treated alike. Then, holding the value fixed, replace the profile a by a common bid equal to the highest bid $a_{(1)}$ with probability $\frac{N-1}{N}$ and to the second-highest bid $a_{(2)}$ with probability $\frac{1}{N}$. The choice of weights is the main step of the argument. Under the original symmetric profile, count how many buyers face which level of competition. Exactly $N - 1$ of the N buyers, that is, everyone except the winner, face a highest competing bid of $a_{(1)}$, and the winner alone faces $a_{(2)}$. The pooled profile therefore reproduces, exactly, the distribution of competition each buyer was already facing. Since a coarse deviation is a single fixed bid, and a fixed bid's profitability depends on rivals only through the highest competing bid, the payoff to every deviation is left unchanged. The price falls from $a_{(1)}$ to $\frac{N-1}{N}a_{(1)} + \frac{1}{N}a_{(2)}$. By (1) the buyers' total payoff rises by the same amount, and by symmetry each buyer's share of it rises. Every obedience constraint therefore has a larger left-hand side and an unchanged right-hand side. Constraints are relaxed, revenue is lower, and the result is still a BCCE.

Remark 2 (Why the weights matter). The construction does not change the law of the highest competing bid. It operates only through the equilibrium payoff. This is why it works for $N > 2$. An alternative, replacing a by a bid drawn uniformly from $\{a_1, \dots, a_N\}$, also lowers revenue, but it strictly raises the probability that a deviator wins whenever $a_{(2)} < \delta \leq a_{(1)}$, and can therefore violate obedience.

Remark 3 (The reduction is specific to first-price pricing). Revenue in the first-price auction is $\mathbb{E}[\max_i a_i]$, which pooling reduces. Under a second-price rule revenue is $\mathbb{E}[a_{(2)}]$, which pooling *raises*. Proposition 1 is therefore a statement about first-price pricing. Section 8 returns to what this implies for the comparison across formats, and Section 6 shows that a modified construction holds with a reserve price.

By the reduction, the remaining problem is to choose the joint distribution of the value v and a single common bid c .

5 Characterization

5.1 The Representative-Bidder Program

Fix an identical play $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$, write c for the common bid and $R = \mathbb{E}[c]$ for expected revenue. Since all N bids coincide, every buyer wins with probability $1/N$ and pays c when they do, so each buyer's equilibrium payoff is

$$K := \frac{m - R}{N}, \quad (3)$$

the per-buyer share of the surplus not collected by the seller. I refer to K as the buyers' *rent*. A buyer who instead commits to the fixed bid δ wins whenever $c < \delta$ and shares when $c = \delta$, so coarse obedience requires, for every $\delta \in [0, \bar{v}]$,

$$K \geq \mathbb{E}\left[(v - \delta)\left(\mathbb{1}\{c < \delta\} + \frac{1}{N}\mathbb{1}\{c = \delta\}\right)\right]. \quad (4)$$

Setting $\delta = 0$ shows $K \geq 0$: revenue can never exceed the expected value.

Two properties of the constraints are used below.

First, lowering revenue relaxes the constraints. By (3) lowering R raises K , and K appears on the left of every constraint (4). The guarantee is the fixed point of this relation, which is why the characterization takes the form of a system rather than a formula.

Second, the constraints at different δ have different roles. Deviations to small δ are deterred by a condition resembling a participation constraint, since a buyer bidding just above zero wins the good exactly in the states where everyone else bids zero. Deviations to larger δ determine the bidding rule. The following definition names the value at which the bidding region begins.

Definition 3 (Threshold and worst-case bidding rule). *Let $\hat{v}$ denote the pooling threshold: in the worst-case equilibrium, buyers bid zero whenever $v < \hat{v}$ and bid a strictly increasing amount above it. Write $H(v)$ for the common bid submitted at value $v \geq \hat{v}$.*

5.2 Main Result

Theorem 1. *Let Assumption 1 hold. The coarse revenue guarantee CRG and the threshold $\hat{v}$ are the unique solution of the system*

$$\frac{m - CRG}{N} = \int_0^{\hat{v}} v f(v) dv, \quad (5)$$

$$CRG = \int_0^{\bar{v}} H(v) f(v) dv, \quad (6)$$

where $H(v) = 0$ for $v < \hat{v}$ and, for $v \geq \hat{v}$,

$$H(v) = \frac{1}{F(v)} \int_{\hat{v}}^v z f(z) dz = \mathbb{E}[z \mid \hat{v} \leq z \leq v] \cdot \frac{F(v) - F(\hat{v})}{F(v)}. \quad (7)$$

The guarantee is strictly positive and can be attained: the identical play distribution with common bid $c = H(v)$ is a BCCE achieving it.

Appendix B contains the proof. There are three main steps. A *lower bound* shows that no identical play BCCE can do better than the system above. This comes from combining the obedience constraint with a rearrangement inequality, rather than from assuming that constraints bind. *Sufficiency* verifies that the candidate rule (7) deters every fixed deviation, including deviations above the top of the bid range. The last step is the *uniqueness* of the solution to (5)–(6). This step is not immediate, since both sides of the system move in the same direction as $\hat{v}$ rises. It follows from a concavity argument given in the appendix.

The fixed-point form. For later use, I record an equivalent form of the system, written in the quantile variable $q = F(v)$. Let

$$\Phi_\mu(q) := \int_0^q F^{-1}(t) dt = \int_0^{F^{-1}(q)} z f(z) dz, \quad q \in [0, 1], \quad (8)$$

the expected value collected on the lowest q quantiles of the prior. It is convex and strictly increasing, with $\Phi_\mu(0) = 0$ and $\Phi_\mu(1) = m$. Define the revenue functional

$$\mathcal{R}_\mu(K) := \int_0^1 \frac{(\Phi_\mu(q) - K)^+}{q} dq, \quad K \in [0, m]. \quad (9)$$

The system (5)–(6) is equivalent to the single equation

$$m - NK = \mathcal{R}_\mu(K) \quad (10)$$

in the rent: the unique solution is $K^* = \int_0^{\hat{v}} v f(v) dv$, the guarantee is $CRG = m - NK^* = \mathcal{R}_\mu(K^*)$, and the worst-case bid is $c = \psi(F(v)) = H(v)$, where

$$\psi(q) := \frac{(\Phi_\mu(q) - K^*)^+}{q}. \quad (11)$$

Appendix B proves the theorem in this form. The functional $\mathcal{R}_\mu$ is also used in the reserve-price analysis of Section 6 and in the convex-order comparison of Section 9.

Reading the solution. Equation (5) is the participation condition. Its right-hand side is the expected value collected on the pooling region, which is exactly what a buyer gets by deviating to a bid just above zero: they win exactly in those states where everyone else has bid nothing. The rent K must be large enough to make that deviation unattractive, and in the worst case it is exactly large enough.

Equation (7) describes the bidding rule above the threshold. The common bid at value v is the expected value conditional on being in the bidding region, shaded by the probability that a win comes from the bidding region rather than from the pooling region. The comparison with the limiting BCE bidding rule $G(v) = \mathbb{E}[z \mid z \leq v]$ shows the source of the gap:

$$G(v) - H(v) = \frac{K}{F(v)} \quad \text{for } v \geq \hat{v}.$$

The coarse bidding rule equals the interim rule minus $K/F(v)$, and the difference between the two guarantees is the integral of this term against f .

Example 2 (Uniform values). Let $\mu = U[0, 1]$, so $m = 1/2$ and $F(v) = v$. Then $H(v) = (v^2 - \hat{v}^2)/(2v)$, and the system (5)–(6) reduces to the single equation

$$\frac{1}{2} - \frac{N\hat{v}^2}{2} = \frac{1}{4} - \frac{\hat{v}^2}{4} + \frac{\hat{v}^2 \ln \hat{v}}{2}.$$

With two buyers the threshold is $\hat{v} \approx 0.6730$ and $CRG \approx 0.0471$. The seller collects less than 5% of the good's expected value. Since $F(\hat{v}) = \hat{v}$ for the uniform distribution, the pooling region has probability 0.673: in the worst case, both buyers bid zero with

probability 0.673, and the good is then sold at a price of zero. Conditional on a positive price being paid, the average price is 0.144. With ten buyers, $CRG \approx 0.1916$ and the pooling probability falls to 0.248.

The pooling probabilities in Example 2 motivate the reserve-price analysis of Section 6.

Proposition 2 (Comparative statics in N). *$CRG(\mu)$ is strictly increasing in N , and the rent K is strictly decreasing in N .*

The guarantee increases in N for a reason specific to coarse rationality. The rent that deters a fixed deviation is a per-buyer quantity: each buyer receives K , and by (3) the total is $NK = m - R$. With more buyers, each one's share of a given amount of uncollected surplus is smaller, so a given level of revenue leaves each buyer with a smaller equilibrium payoff, and the revenue reduction needed to satisfy the constraints is smaller.

5.3 Asymptotic Convergence and Its Rate

Theorem 2. *Under Assumption 1,*

$$\lim_{N \rightarrow \infty} CRG(\mu) = \int_0^{\bar{v}} G(v) f(v) dv = \lim_{N \rightarrow \infty} RG(\mu), \quad G(v) = \mathbb{E}[z \mid z \leq v].$$

The proof is in Appendix C. As N grows, the rent K vanishes at rate $1/N$, the pooling threshold $\hat{v}$ falls to zero, and by the identity $G - H = K/F$ the coarse bidding rule converges pointwise to the interim rule characterized by Bergemann, Brooks and Morris (2017).

A second argument for the same convergence connects the result to the information structure of the benchmark. The worst case in Bergemann, Brooks and Morris is supported by an information structure in which each bidder learns only whether they hold the highest value together with a summary of the remaining values. As N grows, that information structure converges to the value distribution itself, so conditioning on it adds no information, and the two notions of obedience impose the same restriction.

The next question is the rate of convergence. For pure common values, Bergemann, Brooks and Morris (2015), the working paper version of Bergemann, Brooks and Morris (2017), characterize the minimum-revenue BCE: the worst case has a

deterministic winning bid and the corresponding revenue

$$\beta_N(v) = \frac{\int_0^v z dF^{\frac{N-1}{N}}(z)}{F^{\frac{N-1}{N}}(v)}, \quad RG_N(\mu) = \int_0^{\bar{v}} \beta_N(v) f(v) dv, \quad (12)$$

the conditional mean of the value under the tilted distribution $F^{(N-1)/N}$.³ For the uniform prior, $\beta_N(v) = \frac{N-1}{2N-1} v$ and $RG_N = \frac{N-1}{2(2N-1)}$, which equals $1/6$ at $N = 2$ and increases to $1/4$. The interim guarantee therefore converges to the common limit at rate $1/N$, and the next result shows that the coarse guarantee converges strictly more slowly, so the cost of coarseness inherits the slower rate.

Proposition 3 (Rate of convergence). *Suppose f is continuous at 0 with $f(0) \in (0, \infty)$, and let R_∞ denote the limit in Theorem 2. Then $0 \leq R_\infty - RG_N \leq 2\bar{v}/(N-1)$, while*

$$R_\infty - CRG_N = \frac{(m - R_\infty) \ln N}{2N} (1 + o(1)).$$

Consequently the cost of coarseness satisfies

$$RG_N - CRG_N = \frac{(m - R_\infty) \ln N}{2N} (1 + o(1)) :$$

the interim guarantee converges at rate $1/N$, the coarse guarantee at rate $\log N/N$, and the gap between them closes at the slower rate.

The proof is in Appendix C. The hypothesis on f enters at a single point. The gap $R_\infty - CRG_N$ equals the rent K times $\ln(1/q_K)$, up to a term of order $1/N$, and K is of order $1/N$ regardless of the prior. What the hypothesis fixes is how the pooling quantile q_K scales with K . A density that is positive and continuous at zero gives $\Phi_\mu(q) \approx q^2/2f(0)$, hence $q_K \approx \sqrt{2f(0)K}$, and the square root is the factor $\frac{1}{2}$ in the rate.

Table 1 and Figure 1 report the values at matched market sizes. With two bidders the coarse guarantee is under thirty percent of the interim guarantee, and with ten it is about eighty percent.

The inequality in the third row of the table holds for every prior in the class considered and every finite N . The following result shows that the two guarantees

³See Bergemann, Brooks and Morris (2015) for the common value case. In the as-if reading, choose the marginal $F^{1/N}$ so that the highest of N independent values is distributed as F , and then β_N is the textbook symmetric equilibrium bid of that model, the expected highest rival value conditional on winning, and the rival maximum has a distribution $F^{(N-1)/N}$.

N	2	3	5	10	20	50	100
Interim guarantee RG_N	0.1667	0.2000	0.2222	0.2368	0.2436	0.2475	0.2487
Coarse guarantee CRG_N	0.0471	0.0939	0.1453	0.1916	0.2178	0.2353	0.2419
Cost of coarseness $RG_N - CRG_N$	0.1196	0.1061	0.0770	0.0452	0.0258	0.0122	0.0068
CRG_N below RG_N (%)	71.7	53.1	34.6	19.1	10.6	4.9	2.7
Pooling probability $F(\hat{v})$	0.673	0.520	0.377	0.248	0.168	0.103	0.072

Table 1: Interim and coarse revenue guarantees for $\mu = U[0, 1]$, both converging to $1/4$. The interim row is $RG_N = \frac{N-1}{2(2N-1)}$, computed from the [Bergemann, Brooks and Morris \(2017\)](#) characterization via (12). The comparison is at the same market size throughout.

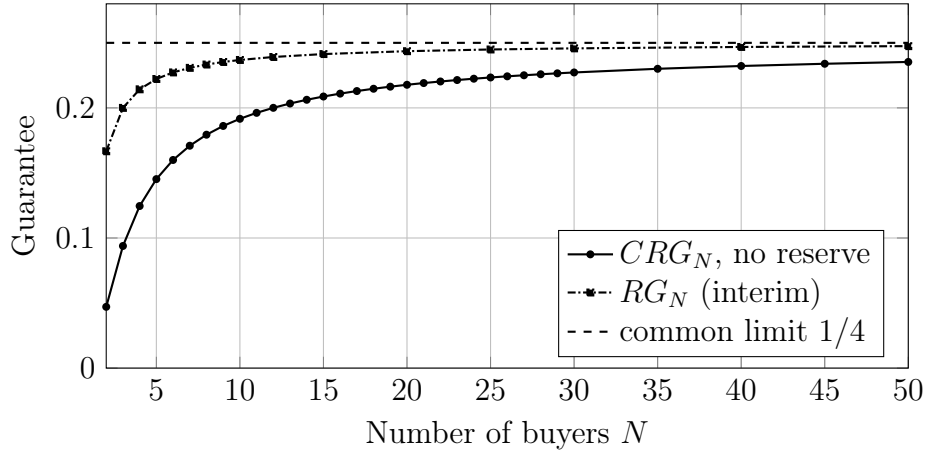


Figure 1: Interim and coarse revenue guarantees for $\mu = U[0, 1]$. The interim guarantee converges to the common limit at rate $1/N$, the coarse guarantee only at rate $\log N/N$, so the vertical distance between the two curves, the cost of coarseness, closes slowly. Section 6 shows that a reserve price recovers much of this gap when N is small.

never coincide at a finite N : the coarse minimum has a rigid structure, identical bids comonotone with the value,⁴ and that structure is incompatible with interim obedience, since a bidder whose recommendation reveals a high value would break the tie.

Proposition 4 (Strict coarseness gap). *Let Assumption 1 hold and suppose the infimum defining $RG_N(\mu)$ is attained, as established by [Bergemann, Brooks and Morris \(2017\)](#). Then*

$$CRG_N(\mu) < RG_N(\mu) \quad \text{for every finite } N.$$

⁴Two random variables are *comonotone* if each is a nondecreasing function of the other along the support. Here, higher values always come with higher bids.

The proof, in Appendix B, has two steps. First, a rigidity lemma: *every* revenue-minimizing BCCE, not only the one constructed in Theorem 1, must have its two highest bids tied almost surely and its winning bid equal to $\psi(F(v))$ almost surely. Second, no distribution with that structure can satisfy interim obedience: a bidder recommended the common bid can profitably raise it slightly, because with positive probability they are tied at the top in a state where the value strictly exceeds the bid. Since Bergemann, Brooks and Morris construct a BCE attaining RG_N , the two minima cannot be equal.

6 Reserve Prices

When N is small, the worst case of Sections 4 and 5 sells the good at a price of zero with high probability. This section shows that a reserve price raises the guarantee substantially if it is large enough. Example 2 shows the mechanism. The worst case is sustained by states in which every buyer bids zero and the buyers receive the good at a price of zero, and the resulting surplus is what satisfies their obedience constraints. A reserve price reduces this surplus. A small reserve reduces it only by the reserve itself, since bidders pool at the reserve and pay it, and only a reserve large enough that the worst case involves withholding eliminates it.

Fix a reserve price $r \in [0, \bar{v})$. The first-price auction with reserve r allocates the good to the highest bidder at the bid price if the highest bid is at least r , breaking ties uniformly, and withholds the good otherwise. Write $CRG_N(r)$ for the resulting coarse revenue guarantee.

Bids below r never win, so without loss of generality every bid is either 0, interpreted as staying out, or at least r . The accounting identity (1) is amended in the following way, since the good is no longer always allocated:

$$\sum_i u_i(v, a) = (v - a_{(1)}) \mathbb{1}\{a_{(1)} \geq r\}. \quad (13)$$

The reduction holds with a modification. The pooling construction of Proposition 1 cannot be applied unchanged. Suppose exactly one buyer clears the reserve, so $a_{(1)} \geq r > a_{(2)}$. Pooling onto $a_{(2)}$ with probability $1/N$ would then withhold the good, and by (13) it would eliminate the buyers' surplus along with the seller's revenue, tightening rather than relaxing obedience. Instead, pool onto $\max\{a_{(2)}, r\}$, so

that the good is still sold, and switch to withholding only in the states where the sale yields negative surplus, namely where the winning bid exceeds the value.

Proposition 5 (Identical Play with a Reserve). *Fix $r \in [0, \bar{v})$. For every BCCE σ of the first-price auction with reserve r there is an exchangeable identical play BCCE γ with $R(\gamma) \leq R(\sigma)$, where the common bid takes values in $\{0\} \cup [r, \bar{v}]$. Consequently $CRG_N(r)$ is the value of the identical play program.*

The proof is in Appendix D. The kernel maps the symmetrized profile to a common bid as follows. If $a_{(1)} < r$, all buyers bid 0. If $a_{(1)} \geq r$ and either $a_{(2)} \geq r$ or $v \geq a_{(1)}$, the common bid is $a_{(1)}$ with probability $\frac{N-1}{N}$ and $\max\{a_{(2)}, r\}$ with probability $\frac{1}{N}$. In the remaining case, where exactly one buyer clears the reserve and the winning bid exceeds the value, the common bid is $a_{(1)}$ with probability $\frac{N-1}{N}$ and 0 with probability $\frac{1}{N}$. The case split ensures that whenever the pooled bid is raised to the reserve, the good is worth at least the reserve, which is why that step lowers rather than raises deviation payoffs. A final normalization then moves the common bid from r to 0 on any remaining states where the good is worth less than the reserve. This lowers revenue and raises the buyers' payoff, and it handles the deviation to exactly r , the one tie not covered by Lemma 2: after the normalization, the value is at least r in every state where the common bid is r , so a deviation to r that ties the pool is unprofitable.

Why a single pool is not the worst case. One might solve the identical play program by assuming that the pooling region of Theorem 1 becomes a no-sale region: bidders bid zero on low values, the good is not sold there, and the surplus that satisfied obedience is removed. That program implies that an arbitrarily small reserve raises the guarantee by as much as one additional bidder, which is false. Under a small reserve, the worst case has bidders pool *at* the reserve. If every buyer bids exactly r on the former pooling region, the good still trades, the winner pays r , and the surplus that sustains obedience falls only by r per state. The seller collects r rather than zero on those states, so a small reserve raises the guarantee, but continuously in r rather than discontinuously.

The two-pool family. The worst case therefore involves up to two pools: a *withholding pool* at zero, where the good is not sold, and a *reserve pool* at exactly r ,

where the good trades at the reserve. By the rearrangement inequality of Lemma 5, the family couples bids and values comonotonically. The withholding pool occupies the lowest s quantiles, the reserve pool the next $q_0 - s$, and above q_0 the common bid is strictly increasing. The family is determined as follows. The cheapest deviation is a bid just above the reserve, which captures both pools outright, so obedience requires $K \geq \Phi_\mu(q_0) - r q_0$. And the bidding rule must start at the reserve, $(\Phi_\mu(q_0) - K)/q_0 \geq r$, which is the same inequality reversed. Hence at the worst case

$$K = \Phi_\mu(q_0) - r q_0. \quad (14)$$

The accounting identity then completes the system: withholding removes value $\Phi_\mu(s)$, so

$$NK = m - \Phi_\mu(s) - R, \quad R = r(q_0 - s) + \int_{q_0}^1 \frac{\Phi_\mu(q) - K}{q} dq, \quad (15)$$

which, given q_0 , determines the withholding share $s \in [0, q_0]$. Let $\widehat{V}_N(r)$ denote the minimum of R over this one-parameter family, subject to $K \geq 0$ and the feasibility condition $\Phi_\mu(q_0) - \Phi_\mu(s) \geq r(q_0 - s)$, which says that the average value in the reserve pool is at least r .

Theorem 3 (Reserve prices). *Fix $r \in [0, m)$ satisfying $rF(r) - \Phi_\mu(F(r)) < (N - 1)(m - r)$, a condition held for every $r \leq m(N - 1)/N$ and in particular for every reserve in the example below. Then*

$$CRG_N(r) = \widehat{V}_N(r),$$

so the two-pool family is exactly the worst case. Moreover the guarantee is continuous at a zero reserve, $\lim_{r \downarrow 0} CRG_N(r) = CRG_N(0)$.⁵ For $r \geq m$ the guarantee is zero, since universal rejection becomes an equilibrium.

The proof is in Appendix D and uses two steps beyond those of Theorem 1. The first is a comonotonization lemma, which plays the role at the level of couplings that the pooling kernel plays at the level of bid profiles in Proposition 1. Rearranging the bid-value coupling into its comonotone form, holding the bid marginal fixed, leaves revenue unchanged, raises the buyers' rent, and lowers every deviation payoff,

⁵The characterization was also confirmed by directly solving a discretized version of the coarse program as a linear program, with neither identical play nor comonotonicity imposed. The linear program reproduces $\widehat{V}_N(r)$ at every reserve tested.

r	0.01	0.05	0.10	0.148	0.20
$CRG_2(r)$	0.0552	0.0812	0.1000	0.1056	0.1038
withholding share s	0.051	0.253	0.464	0.621	0.664
pool boundary q_0	0.676	0.674	0.650	0.621	0.664

Table 2: The worst case across reserve levels, $\mu = U[0, 1]$, $N = 2$. The reserve pool has mass $q_0 - s$. It contains most of the pooled mass at small reserves, shrinks as the reserve grows, and vanishes at the revenue-maximizing reserve, $r \approx 0.148$, beyond which the worst case relies on withholding alone. Values computed from (14)–(15).

so the worst case couples bids and values comonotonically. The second ingredient reduces the remaining problem to a one-dimensional concave fixed point for each withholding share, of exactly the form solved in Theorem 1. The largest root bounds the rent that any equilibrium with that share can sustain, the bound is attained by the corresponding family member, and shares whose pools lie below the reserve are dominated, which removes the tie constraint.

Example 3 (Uniform values with a reserve). Return to the setting of Example 2, where $\mu = U[0, 1]$, $N = 2$, and without a reserve $CRG_2 \approx 0.047$, sustained by both buyers bidding zero on the lowest two-thirds of values and receiving the good at a price of zero. Table 2 traces the worst case as the reserve grows. At $r = 0.01$ nearly the entire former pooling region bids r , almost nothing is withheld, and the guarantee rises continuously from $CRG_2(0)$, with slope roughly 0.8. As the reserve grows, pooling at the reserve becomes costly for the buyers and the worst case shifts toward withholding, so the share s rises. The guarantee is maximized at $r \approx 0.148$, which is the point where the reserve pool vanishes. At a smaller reserve part of the pooling region still bids r , and at a larger one the probability of sale falls without further tightening obedience. At that reserve $CRG_2 \approx 0.106$, more than double the guarantee without one, and the gain accrues gradually, since at smaller reserves part of the pooling region bids r : reaching half of it requires $r \approx 0.041$. With two coarse bidders, the guarantee without a reserve is less than half of the guarantee at the optimal reserve.

7 Beyond Common Values: Symmetric Priors

The analysis so far, with and without a reserve, fixes the pure common value model. Since the interim characterization of [Bergemann, Brooks and Morris \(2017\)](#) covers every symmetric and arbitrarily correlated prior, this section asks how much of the analysis extends to values that may differ across buyers. Let π be an exchangeable prior on V^N with common per-buyer marginal μ , let buyer i 's payoff be $q_i(a)(v_i - a_i)$, and define BCCE exactly as in [Lemma 1](#), now over joint laws of $(v_1, \dots, v_N, a_1, \dots, a_N)$ with value marginal π . Pure common values are the coupling $v_1 = \dots = v_N$, and independent private values are the opposite extreme. Write $CRG(\pi)$ for the infimum of revenue over all BCCE under π , and $\mathcal{V}_N(\pi)$ for the infimum over *identical play* BCCE. Identical play distributions form a subset of all BCCE, so $CRG(\pi) \leq \mathcal{V}_N(\pi)$, with equality under common values by [Proposition 1](#). This raises the question of which coupling is the worst within identical play, and the further question of whether identical play is still without loss. The answers are common values and no.

7.1 Common Values Are the Worst Coupling Within Identical Play

Under BCE, the correlation structure of the prior is irrelevant to the seller's problem, in the sense that their method handles every coupling. Under BCCE it matters, and within identical play the common value coupling is the worst one.

Proposition 6 (Common values are the worst coupling within identical play). *For every exchangeable prior π with per-buyer marginal μ ,*

$$\mathcal{V}_N(\pi) \geq CRG(\mu),$$

and the bound is attained if and only if the buyers' values coincide wherever bids are positive: $v_1 = \dots = v_N$ holds π -almost surely on the event $\{\max_i v_i > \hat{v}\}$, where $\hat{v}$ is the threshold of [Theorem 1](#). In particular the pure common value coupling attains the bound.

The proof is in [Appendix E](#). The lower bound holds because the rearrangement inequality of [Lemma 5](#) uses only the marginal distribution of each buyer's value. Whatever the coupling, the value a deviator collects on the event $\{c \leq \delta\}$ is at least the

value of the lowest $\Pr(c \leq \delta)$ quantiles of μ . Attaining the bound, however, requires that single event to be a lower-contour set of *every* buyer's value simultaneously, at every binding deviation. A single common bid can only be aligned with one value at a time. When the values agree above the threshold, one comonotone bid constrains all N buyers at the same time, and the constraints below the threshold never bind, so the coupling within the pooling region is unrestricted. When they do not agree, some buyer's constraint must be slack and revenue is higher.

The contrast with the interim analysis comes from how each solution concept constrains a bidder. Under BCE, obedience is checked conditional on the bidder's own recommendation, so nature can correlate each bidder's signal with that bidder's own value. This gives nature N separate instruments, one per bidder, and it is why the [Bergemann, Brooks and Morris](#) worst case handles arbitrary couplings and is even allocatively efficient. Under BCCE, a bidder's own bid appears nowhere in their own deviation payoff $\mathbb{E}[(v_i - \delta) \mathbb{1}\{\max_{j \neq i} a_j < \delta\}]$, so the only thing that constrains buyer i is the correlation between v_i and the *other* buyers' bids. Within identical play that is a single common bid, which also determines the allocation, and it can be aligned with all N values at once exactly when the values are the same.

7.2 Identical Play Is Not Without Loss

Proposition 6 compares couplings inside the identical play program. Whether that program gives the guarantee away from common values depends on whether the reduction of Proposition 1 holds, and it does not in general. The symmetrization half still holds: permuting values and bids together preserves an exchangeable prior, maps each buyer's coarse constraint to another buyer's, and leaves revenue unchanged (Lemma 15). The pooling half fails. Apply the kernel of Lemma 4 to an exchangeable BCCE $\hat{\sigma}$, and write v_{top} for the value of the buyer who wins under $\hat{\sigma}$, $v_{-\text{top}}$ for the average value of the other $N - 1$ buyers, and $\bar{v}$ for the average of all N . The two comparisons in the proof of Lemma 4 now carry correction terms (Lemma 16):

$$\begin{aligned} K_\gamma - K_{\hat{\sigma}} &= \frac{1}{N} \mathbb{E}[\bar{v} - v_{\text{top}}] + \frac{1}{N^2} \mathbb{E}[a_{(1)} - a_{(2)}], \\ D_\gamma(\delta) - D_{\hat{\sigma}}(\delta) &= \frac{N-1}{N^2} \mathbb{E}[(v_{-\text{top}} - v_{\text{top}}) \mathbb{1}\{a_{(2)} < \delta \leq a_{(1)}\}], \end{aligned} \tag{16}$$

where K is the rent and $D(\delta)$ the payoff from the fixed deviation δ . Under common values both corrections vanish, which is the content of Lemma 4. In general the

first says that identical play allocates the good to a uniformly random buyer instead of to the top bidder, so if the top bidder has above-average value the pooled rent falls, and the second says that a deviator who outbids the others but not the top bidder is, by exchangeability, someone other than the top bidder. Pooling therefore works when $\hat{\sigma}$ allocates against value and fails when it allocates efficiently, and with heterogeneous values the adversary lowers revenue by allocating efficiently: the good goes to the buyer with the highest value, the buyers' total surplus rises, and the additional surplus relaxes the obedience constraints, which permits lower bids.

In the first-price auction the adversary can do this at an arbitrarily small cost in revenue. Fix a common price and give the buyer with the highest value a bid premium of ε above it. The premium decides the allocation and adds only ε to revenue, while every coarse deviation still faces, up to ε , the common price as its highest competing bid. The next result shows that this is the generalization of Theorem 1. Let $m_N := \mathbb{E}_\pi[\max_i v_i]$ denote the expected highest value, so that $m \leq m_N \leq Nm$, and let $\tilde{K}$ be the unique solution in $(0, m]$ of

$$m_N - NK = \mathcal{R}_\mu(K), \quad (17)$$

which exists by the argument of Lemma 6 with m_N in place of m . Write $\tilde{q} := \Phi_\mu^{-1}(\tilde{K})$ and $\tilde{v} := F^{-1}(\tilde{q})$ for the associated pooling quantile and threshold.

Theorem 4 (Symmetric priors). *Let π be exchangeable with per-buyer marginal μ satisfying Assumption 1. Then*

$$CRG(\pi) \geq m_N - N\tilde{K} = \mathcal{R}_\mu(\tilde{K}).$$

The inequality holds with equality if $v_1 = \dots = v_N$ holds π -almost surely on $\{\max_i v_i > \tilde{v}\}$ and, with positive probability, at least two buyers have strictly positive values on $\{\max_i v_i \leq \tilde{v}\}$. It is strict if $v_1 = \dots = v_N$ fails with positive probability on $\{\max_i v_i > \tilde{v}\}$.

The proof is in Appendix E. The lower bound is the chain of Theorem 1 with two substitutions: the rearrangement inequality is applied to the pair (v_i, M_i) of a buyer's own value and highest competing bid, which uses only the marginal μ , and the accounting identity bounds the buyers' total surplus by the efficient surplus $m_N - R$ rather than by $m - R$. The attainment construction was described above. On the

pooling event $\{\max_i v_i \leq \tilde{v}\}$ the highest-value buyer bids ε and the others bid zero, above it all buyers bid a common function of the common value, and as $\varepsilon \downarrow 0$ revenue converges to the bound. The non-degeneracy condition in the equality case rules out the case in which at most one buyer ever has a positive value in the pool, where the premium cannot raise the buyers' total surplus. The strictness claim is the stability argument of Proposition 6 applied buyer by buyer to the highest competing bid.

One corollary is that, in the equality case, the guarantee lies strictly below the common value guarantee whenever the values differ with positive probability on the pooling event, since then $m_N > m$, the fixed point (17) lies above the fixed point (10), and $\mathcal{R}_\mu$ is strictly decreasing:

$$CRG(\pi) = \mathcal{R}_\mu(\tilde{K}) < \mathcal{R}_\mu(K^*) = CRG(\mu) \leq \mathcal{V}_N(\pi).$$

Common values are thus the worst coupling for the seller within identical play but not overall. The worst couplings agree at high values, where a single common bid must discipline every buyer, and disperse at low values, where differing values cost nothing and efficient allocation generates rent. One qualification is that the infimum in the equality case is approached along a sequence in which the top two bids differ by a vanishing premium and is not attained: for $N = 2$, attainment would require the two bids to tie almost surely and the winner to hold the higher value almost surely, which the uniform tie-break cannot arrange unless the values coincide.

8 Ranking and Equivalence Across Auctions

I return to common values and compare the first-price auction with other formats, following Bergemann, Brooks and Morris (2019). Following their framework, a *standard auction* is one that allocates the good to the highest bidder, breaking ties uniformly, and admits an equilibrium in monotonic pure strategies in the symmetric independent private values model. Losers pay nothing, and the winner pays $p(a)$, which I take to satisfy $p(a) \leq a_{(1)}$ and $p(b, \dots, b) = b$: the winner never pays more than their bid, and tied bidders pay their common bid. Let $\mathcal{M}$ denote this class. Within it, the pricing rules that matter are defined by what happens on profiles in which the $N - 1$ losing bids coincide. A standard auction has *own-bid pricing* if the winner pays $a_{(1)}$ whenever the other bids coincide, as in the first-price and Dutch auctions, and

competing-bid pricing if the winner pays $a_{(2)}$ whenever the other bids coincide, as in the second-price and English auctions.

The comparison uses the following two facts, combined as in [Bergemann, Brooks and Morris](#). The first is that identical play is an outcome every standard auction can produce, at the same revenue. The second is that formats differ only in what a single deviator pays after outbidding the others.

Proposition 7 (Identical play is format-free). *Fix any identical play distribution with common bid c . In every standard auction the good is allocated uniformly among the N buyers, the winner pays c , and expected revenue is $\mathbb{E}[c]$. In every standard auction with own-bid pricing, moreover, the identical play BCCE are exactly those of the first-price auction, so the coarse revenue guarantee restricted to identical play equals $CRG(\text{FPA})$ in each of them.*

Proof. All bids are tied at c , so the allocation is the uniform tie-break and the price is $p(c, \dots, c) = c$. Under own-bid pricing, a buyer who deviates from an identical play distribution to the fixed bid δ loses if $\delta < c$, ties at price c if $\delta = c$, and wins at price $p(\delta, c, \dots, c) = \delta$ if $\delta > c$, exactly as in the first-price auction, and equilibrium payoffs coincide by the first part. So the coarse constraints (2) are the same in both formats, and the sets of identical play BCCE coincide. $\square$

Proposition 8 (Coarse revenue guarantee ranking). *The first-price auction has the highest coarse revenue guarantee among standard auctions with own-bid or competing-bid pricing. For every $M \in \mathcal{M}$ with own-bid pricing, $CRG(M) \leq CRG(\text{FPA})$, and for every $M \in \mathcal{M}$ with competing-bid pricing, in particular the second-price and English auctions,*

$$CRG(M) = 0 < CRG(\text{FPA}).$$

Proof. Under own-bid pricing, the identical play BCCE attaining $CRG(\text{FPA})$ in Theorem 1 is a BCCE of M with the same revenue, by Proposition 7, so $CRG(M) \leq CRG(\text{FPA})$. Under competing-bid pricing, consider the distribution in which one designated buyer bids $\bar{v}$ and all others bid zero, independently of the value. Revenue is zero. The designated buyer earns m and cannot improve, since any positive bid still wins at the price $p(\delta, 0, \dots, 0) = 0$. Any other buyer must bid $\bar{v}$ to contest the good, which ties the top bid and, by $p(b, \dots, b) = b$, means winning with some probability $\lambda \in (0, 1]$ at price $\bar{v}$, for a payoff of $\lambda(m - \bar{v}) \leq 0$, and the conclusion does not depend

on how ties are broken. This is a BCCE with zero revenue, so $CRG(M) = 0$, and $CRG(\text{FPA}) > 0$ by Theorem 1. $\square$

The same argument underlies the results of [Bergemann, Brooks and Morris \(2019\)](#). They show that all standard auctions have the same revenue guarantee once attention is restricted to symmetric affiliated models and monotonic pure-strategy equilibria, and that outside that restriction the first-price auction has the highest guarantee, because the other formats admit asymmetric equilibria the restriction excludes. Here the restriction that gives equivalence is identical play. The outcome that attains the first-price guarantee is available in every standard auction, and in every format where a deviator who outbids the others pays their own bid it is also an equilibrium, so no such format can do better. Where the deviator pays the others' bid instead, outbidding them does not change the price paid, pooling at a low price cannot be sustained, and the comparison is decided by an asymmetric equilibrium in which one buyer bids the maximum and the rest bid nothing. That equilibrium is a complete-information Nash equilibrium, hence a BCE and a BCCE, and its zero revenue is well known. What the coarse analysis adds to the ranking is therefore not the second-price auction's zero but the positive side, the value of $CRG(\text{FPA})$ from Theorem 1, together with the conclusion that a seller designing for no-regret bidders is led to the same format as a seller designing for Bayesian bidders, even though the worst cases differ.

9 Unknown Value Distribution

I next address a seller who knows the expected value $E = \mathbb{E}[v]$ and the upper bound $\bar{v}$, but not the shape of the distribution. This requires dropping the density assumption. The comparison ranges over every distribution on $[0, \bar{v}]$ with the given mean, and the worst case among them is atomic, so the density maintained by Assumption 1 must first be dropped. The characterization holds in its fixed-point form.

Theorem 5 (Beyond the density). *Let μ be any non-degenerate distribution on $[0, \bar{v}]$, and read the definitions (8), (9), and (11) with the quantile function $F^{-1}(t) = \inf\{x : F(x) \geq t\}$ in place of the inverse. Then (10) has a unique solution $K^* \in (0, m)$, and $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*) > 0$. The infimum is attained by the identical play distribution $v = F^{-1}(U)$, $c = \psi(U)$, with U uniform on $[0, 1]$.*

The uniform variable is the rank of the value. Off atoms $U = F(v)$, so the coupling reduces to the deterministic bid $H(v) = \psi(F(v))$ of Theorem 1. Conditional on an atom of μ , however, U is uniform on the corresponding quantile interval, so the common bid is randomized continuously across it. The randomization is required by the constraints. Minimality requires the obedience constraint to bind at every deviation level, and a deviation to $\psi(q)$ with q inside an atom's interval must capture exactly a fraction of the atom, which a single common bid cannot arrange. If the whole atom bid one number, a deviation just above it would capture the entire mass at once and earn strictly more than the rent.

The proof, in Appendix F, checks that the argument of Appendix B holds with atoms. The identical play reduction of Proposition 1 does not use the density. Three steps of the characterization use it, and each is modified: the rearrangement inequality holds for arbitrary priors once its equality clause allows an atom to be split, the concavity lemma behind uniqueness does not use the density, and the attaining coupling randomizes the common bid at any atom of the prior, in the manner just described.

The relevant comparison is then the convex order: $\mu \preceq_{\text{cx}} \nu$, read as “ ν is riskier than μ ,” means that μ and ν have the same mean and that ν can be obtained from μ by a sequence of mean-preserving spreads.

Proposition 9 (Convex order monotonicity). *If $\mu \preceq_{\text{cx}} \nu$ then $\text{CRG}(\nu) \leq \text{CRG}(\mu)$, for every N . Consequently, for every N ,*

$$\min_{\mu: \text{supp } \mu \subseteq [0, \bar{v}], \mathbb{E}_\mu[v] = E} \text{CRG}(\mu) = \text{CRG}(\underline{\mu}),$$

where $\underline{\mu}$ is the binary distribution with $\underline{\mu}(\{0\}) = 1 - E/\bar{v}$ and $\underline{\mu}(\{\bar{v}\}) = E/\bar{v}$.

The proof is in Appendix F. What sustains a low price in the worst case is that a buyer who deviates to a slightly higher bid wins mostly in states where the value is low, so the deviation is unprofitable despite the low price. This relies on the bid being increasing in the value. A mean-preserving spread moves probability mass toward the extremes and weakens this relation, so a larger rent is needed to satisfy the obedience constraints, and revenue falls. Since the binary distribution on $\{0, \bar{v}\}$ is the riskiest distribution on $[0, \bar{v}]$ with a given mean, it is the seller's worst case.

Corollary 1 (The binary guarantee). *Let $p_0 = 1 - E/\bar{v}$. For finite N , $\text{CRG}(\underline{\mu}) = E - N\bar{v}(q^* - p_0)$, where $q^* \in (p_0, 1)$ is the unique solution of $(N - 1)(p_0 - q) = q \ln q$.*

As $N \rightarrow \infty$,

$$CRG(\underline{\mu}) \longrightarrow E + (\bar{v} - E) \ln \left(1 - \frac{E}{\bar{v}}\right). \quad (18)$$

Stating the result through the convex order, rather than through the extreme point alone, has three advantages. It holds at every N , not only in the limit. It covers distributions with atoms, which matters because the minimizer is itself atomic. And it gives a full comparative static rather than just the worst case, so a seller with partial information about dispersion can locate their guarantee between the extremes. For $E = 1/2$ and $\bar{v} = 1$, the ordering $U[0, 1] \preceq_{\text{cx}} \underline{\mu}$ corresponds to the reverse ordering of guarantees: 0.047 against 0.040 at $N = 2$, and 0.250 against 0.153 in the limit. A more dispersed prior lowers the guarantee.

10 Conclusion

This paper characterizes the worst-case revenue of auctions when bidders are rational only in the coarse sense that no single fixed bid would have served them better on average. No-regret learning leads to this restriction, and it is the restriction that best describes algorithmic bidders. In the first-price auction the worst case is an identical play equilibrium, in which bidders do not use their private information and all submit the same bid, and this reduction yields a closed-form characterization of the guarantee. The guarantee is strictly positive, increasing in the number of bidders, and strictly below the interim benchmark of [Bergemann, Brooks and Morris \(2017\)](#) at every finite market size. The two converge, but the interim guarantee approaches the common limit at rate $1/N$ while the coarse guarantee approaches it only at rate $\log N/N$, so the gap remains large for moderate N . Among symmetric priors with a given marginal, common values are the worst coupling within identical play, but identical play is not without loss once values differ: the adversary can allocate efficiently at a common price, and the guarantee becomes the same fixed point with the mean replaced by the expected highest value, which lies strictly below the common value guarantee whenever values differ in the pooling region.

The worst case is sustained by states in which every bidder bids zero and the buyers receive the good at a price of zero, and the resulting surplus is what satisfies their obedience constraints. A reserve price reduces this surplus, but a small reserve reduces it only slightly: coarse bidders pool at the reserve and pay it, the surplus

falls only by the reserve, and the guarantee is continuous at a zero reserve. The revenue-maximizing reserve is the smallest reserve at which the reserve pool vanishes and the worst case involves withholding only, and it more than doubles the guarantee with two bidders. Across formats, the first-price auction has the highest guarantee: identical play is available at the same revenue in every standard auction, and where a deviator pays the others' bid rather than their own, the guarantee is zero. Finally, the guarantee is monotone in the convex order, so a seller who knows only the mean should benchmark against the binary distribution.

Several questions remain. Away from common values the guarantee is characterized exactly only for priors that agree on the bidding region. For priors that do not satisfy this condition, the coarse revenue guarantee remains open. Whether the first-price auction maximizes the coarse revenue guarantee over all standard auctions, as it maximizes the interim guarantee in [Bergemann, Brooks and Morris \(2019\)](#), is also open, since the ranking proved here covers the formats in which a deviator who outbids tied opponents pays either their own bid or the opponents' bid. And the optimal mechanism against coarsely rational bidders, of which the reserve price studied here is only one simple instrument, remains open as well.

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A Proof of the Identical Play Reduction

Throughout, for an action profile $a \in A$ let $a_{(1)} \geq a_{(2)} \geq \dots \geq a_{(N)}$ denote the order statistics. The map $a \mapsto (a_{(1)}, a_{(2)})$ is continuous and hence measurable. For $c \in [0, \bar{v}]$ write $\mathbf{c} = (c, \dots, c)$.

Lemma 3 (Symmetrization). *For $\sigma \in \mathcal{BCC}\mathcal{E}(\mu)$ let $\hat{\sigma} := \frac{1}{N!} \sum_{\pi} \sigma^{\pi}$, where σ^{π} is the image of σ under $(v, a) \mapsto (v, a_{\pi})$ with $(a_{\pi})_i = a_{\pi(i)}$. Then $\hat{\sigma} \in \mathcal{BCC}\mathcal{E}(\mu)$ is exchangeable and $R(\hat{\sigma}) = R(\sigma)$.*

Proof. Permuting action coordinates leaves the value marginal equal to μ , since the value is common. Since $u_i(v, a_{\pi}) = u_{\pi(i)}(v, a)$ and a coarse deviation is a constant, buyer i 's constraint under σ^{π} is buyer $\pi(i)$'s constraint under σ , so each σ^{π} lies in $\mathcal{BCC}\mathcal{E}(\mu)$. The constraints (2) are linear in σ , so $\mathcal{BCC}\mathcal{E}(\mu)$ is convex and the average $\hat{\sigma}$ belongs to it. Finally $\max_i a_i$ is permutation invariant and R is linear, so revenue is unchanged. $\square$

Lemma 4 (Pooling). *Let $\hat{\sigma} \in \mathcal{BCC}\mathcal{E}(\mu)$ be exchangeable and let γ be the image of $\hat{\sigma}$ under the Markov kernel*

$$\Gamma((v, a), \cdot) = \delta_{\{v\}} \otimes \left[\frac{N-1}{N} \delta_{\{\mathbf{a}_{(1)}\}} + \frac{1}{N} \delta_{\{\mathbf{a}_{(2)}\}} \right].$$

Then $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$ is exchangeable identical play, and

$$R(\gamma) = \mathbb{E}_{\hat{\sigma}} \left[\frac{N-1}{N} a_{(1)} + \frac{1}{N} a_{(2)} \right] \leq R(\hat{\sigma}),$$

strictly unless $a_{(1)} = a_{(2)}$ holds $\hat{\sigma}$ -almost surely.

Proof. The kernel places all mass on identical profiles and leaves v unchanged, so γ is identical play, exchangeable, and has value marginal μ . Revenue under γ is the common bid, which gives the displayed expression. Since $a_{(2)} \leq a_{(1)}$ and $R(\hat{\sigma}) = \mathbb{E}_{\hat{\sigma}}[a_{(1)}]$, revenue weakly falls, strictly unless the top two bids coincide almost surely.

Step 1: the law of the highest competing bid is unchanged. Fix a buyer i and let $M_i := \max_{j \neq i} a_j$. Condition on (v, a) under $\hat{\sigma}$ and average across the N buyers, which exchangeability permits. If $a_{(1)} > a_{(2)}$, exactly one buyer, the unique top bidder, faces $M_i = a_{(2)}$, and the remaining $N - 1$ face $M_i = a_{(1)}$. If $a_{(1)} = a_{(2)}$, every buyer faces $M_i = a_{(1)}$. In both cases the conditional law of M_i is $\frac{N-1}{N} \delta_{\{a_{(1)}\}} + \frac{1}{N} \delta_{\{a_{(2)}\}}$, which by

construction of Γ is exactly the conditional law of the common bid c under γ . Hence (v, M_i) and (v, c) have the same joint law.

Step 2: deviation payoffs. Let D be the set of δ that are not atoms of the law of c . Its complement is countable, so D is dense. Fix $\delta \in D$. By Step 1, δ is not an atom of the law of M_i either, so ties occur with probability zero under both distributions and

$$\mathbb{E}_\gamma[(v - \delta)\mathbb{1}\{c < \delta\}] = \mathbb{E}_{\hat{\sigma}}[(v - \delta)\mathbb{1}\{M_i < \delta\}] = (\text{buyer } i\text{'s deviation payoff under } \hat{\sigma}).$$

Step 3: equilibrium payoffs weakly rise. The good is always allocated, so by (1) the buyers' total payoff is $v - a_{(1)}$ under $\hat{\sigma}$ and $v - c$ under γ . Both distributions are exchangeable, so each buyer's expected payoff is $1/N$ of the expected total:

$$\mathbb{E}_{\hat{\sigma}}[u_i] = \frac{1}{N}\mathbb{E}_{\hat{\sigma}}[v - a_{(1)}] \leq \frac{1}{N}\mathbb{E}_{\hat{\sigma}}\left[v - \frac{N-1}{N}a_{(1)} - \frac{1}{N}a_{(2)}\right] = \mathbb{E}_\gamma[u_i] =: K_\gamma.$$

Combining, for every $\delta \in D$ we have $K_\gamma \geq \mathbb{E}_{\hat{\sigma}}[u_i] \geq \mathbb{E}_{\hat{\sigma}}[(v - \delta)\mathbb{1}\{M_i < \delta\}] = \mathbb{E}_\gamma[(v - \delta)\mathbb{1}\{c < \delta\}]$. Lemma 2 extends the inequality to every $\delta \in [0, \bar{v}]$, ties included. Hence $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$. $\square$

Proof of Proposition 1. Compose Lemmas 3 and 4. The statement is a map from an arbitrary BCCE to an identical play BCCE with weakly lower revenue, so no minimizer need be presumed to exist, and the equality of the two infima follows. $\square$

B Proof of Theorem 1

Assumption 1 is in force, so F is continuous and strictly increasing on $[0, \bar{v}]$ with inverse F^{-1} , and Φ_μ , $\mathcal{R}_\mu$, and ψ are as in (8), (9), and (11). I prove the theorem in its fixed-point form: the equation (10) has a unique solution $K^* \in (0, m)$, no identical play BCCE, and hence by Proposition 1 no BCCE at all, yields revenue below $m - NK^*$, and the comonotone coupling $a = \psi(F(v))$ attains it, so that $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*) > 0$. Theorem 1 follows by the substitution $q = F(v)$: the threshold is defined through $\Phi_\mu(F(\hat{v})) = K^*$, that is, $\int_0^{\hat{v}} z f(z) dz = K^*$. Then $\psi(F(v)) = H(v)$ as in (7) and (10) becomes (5)–(6).

Throughout, $v \sim \mu$, $R = \mathbb{E}[c]$, $K = (m - R)/N$, and $A(q) = \inf\{x : \Pr(c \leq x) \geq q\}$ is the quantile function of the bid, so $R = \int_0^1 A(q) dq$.

Lemma 5 (Rearrangement bound). *For any event B with $\Pr(B) = q$, $\mathbb{E}[v\mathbb{1}_B] \geq \Phi_\mu(q)$, with equality if and only if B coincides with the lower-contour set $\{v \leq F^{-1}(q)\}$ up to null sets.*

Proof. This is the Hardy–Littlewood rearrangement inequality (Hardy, Littlewood and Pólya, 1934): among events of probability q , the expectation $\mathbb{E}[v\mathbb{1}_B]$ is smallest when B is placed on the lowest realizations of v , which gives $\int_0^q F^{-1}(t)dt$. $\square$

By Lemma 5, monotonicity of the worst-case bidding rule is derived rather than assumed. The bound below applies to every identical play BCCE, monotone or not, and is then shown to be attained by the comonotone coupling.

Step 1: the lower bound. Fix any identical play $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$. For $\delta \in [0, \bar{v})$ and $\delta' \in (\delta, \bar{v}]$, (4) implies $K \geq \mathbb{E}[(v - \delta')\mathbb{1}\{c < \delta'\}]$. Letting $\delta' \downarrow \delta$ and applying dominated convergence,

$$K \geq \mathbb{E}[(v - \delta)\mathbb{1}\{c \leq \delta\}] = \mathbb{E}[v\mathbb{1}\{c \leq \delta\}] - \delta F_c(\delta) \geq \Phi_\mu(F_c(\delta)) - \delta F_c(\delta), \quad (19)$$

where the last step is Lemma 5. Fix $q \in (0, 1)$ and set $\delta = A(q)$. Right-continuity of F_c gives $F_c(A(q)) \geq q$. The map $q \mapsto \frac{\Phi_\mu(q) - K}{q}$ is nondecreasing, since its derivative has the sign of

$$F^{-1}(q)q - \Phi_\mu(q) + K = \int_0^q (F^{-1}(q) - F^{-1}(t)) dt + K \geq 0.$$

Hence (19) evaluated at $\delta = A(q)$ yields $A(q) \geq \frac{\Phi_\mu(F_c(A(q))) - K}{F_c(A(q))} \geq \frac{\Phi_\mu(q) - K}{q}$, and since $A \geq 0$,

$$R = \int_0^1 A(q) dq \geq \int_0^1 \frac{(\Phi_\mu(q) - K)^+}{q} dq = \mathcal{R}_\mu(K). \quad (20)$$

Together with $R = m - NK$ this gives $g_N(K) \geq 0$, where $g_N(K) := m - NK - \mathcal{R}_\mu(K)$.

Lemma 6 (Concavity and uniqueness). *$\mathcal{R}_\mu$ is convex, nonincreasing and differentiable on $(0, m)$ with $\mathcal{R}'_\mu(K) = \ln q_K$, where $q_K := \Phi_\mu^{-1}(K)$ is the unique solution of $\Phi_\mu(q_K) = K$. Hence g_N is concave on $[0, m]$, with $g_N(0) = m - \int_0^1 \frac{\Phi_\mu(q)}{q} dq > 0$ and $g_N(m) = -(N-1)m < 0$, so g_N has a unique zero $K^* \in (0, m)$, is positive on $[0, K^*)$ and negative on $(K^*, m]$.*

Proof. Φ_μ is continuous and strictly increasing, so $\{q : \Phi_\mu(q) > K\} = (q_K, 1]$ and $\mathcal{R}_\mu(K) = \int_{q_K}^1 \frac{\Phi_\mu(q) - K}{q} dq$. The integrand vanishes at q_K , so differentiating under the integral gives $\mathcal{R}'_\mu(K) = -\int_{q_K}^1 \frac{dq}{q} = \ln q_K \leq 0$. Since q_K is nondecreasing in K , $\mathcal{R}'_\mu$ is nondecreasing, so $\mathcal{R}_\mu$ is convex and g_N concave. For the endpoints, $\frac{\Phi_\mu(q)}{q} = \mathbb{E}[v \mid v \leq F^{-1}(q)] < m$ for $q < 1$, since μ is non-degenerate under Assumption 1, giving $g_N(0) > 0$, and $\mathcal{R}_\mu(m) = 0$. A concave function positive at the left endpoint and negative at the right endpoint has exactly one zero and crosses from above. $\square$

By Lemma 6, $g_N(K) \geq 0$ forces $K \leq K^*$ and therefore $R = m - NK \geq m - NK^*$, which is the required lower bound.

Step 2: attainment. Consider the identical play distribution with common bid $c = \psi(F(v))$. The value marginal is μ by construction, and since $F(v)$ is uniform on $[0, 1]$, revenue is $\mathbb{E}[\psi(F(v))] = \int_0^1 \psi = \mathcal{R}_\mu(K^*) = m - NK^*$ and the induced rent is exactly K^* . It remains to check obedience. Because $K^* > 0$, for $q > q_{K^*}$

$$\psi'(q) = \frac{F^{-1}(q)q - \Phi_\mu(q) + K^*}{q^2} = \frac{\int_0^q (F^{-1}(q) - F^{-1}(t))dt + K^*}{q^2} > 0,$$

so ψ is continuous, zero on $[0, q_{K^*}]$, and strictly increasing on $[q_{K^*}, 1]$ with $\psi(1) = m - K^*$. The bid therefore has a single atom, of size q_{K^*} , at zero, and every $\delta \in (0, m - K^*)$ equals $\psi(q)$ for exactly one q . Three cases cover all deviations.

- $\delta = \psi(q)$ for some $q \in (q_{K^*}, 1]$. Then $\Pr(c < \delta) = \Pr(F(v) < q) = q$ and $\{c < \delta\} = \{v < F^{-1}(q)\}$, so

$$\mathbb{E}[(v - \delta)\mathbb{1}\{c < \delta\}] = \Phi_\mu(q) - \psi(q)q = \Phi_\mu(q) - (\Phi_\mu(q) - K^*) = K^*,$$

and there is no tie term since $\Pr(c = \delta) = 0$. The constraint binds exactly.

- $\delta = 0$. The deviator ties with the atom and wins with probability $1/N$, earning $\frac{1}{N}\mathbb{E}[v\mathbb{1}\{c = 0\}] = \frac{\Phi_\mu(q_{K^*})}{N} = \frac{K^*}{N} < K^*$.
- $\delta > \psi(1) = m - K^*$. The deviator always wins and earns $m - \delta < K^*$.

Hence the coupling is a BCCE, and $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*)$, which is strictly positive because $K^* < m$. Note also that $\psi(q) \leq F^{-1}(q)$ pointwise, since $\Phi_\mu(q) \leq qF^{-1}(q)$: the worst-case bid never exceeds the value, as required, since a bidder pooled above the value could profitably commit to zero. $\square$

Uniqueness in the threshold variable. Lemma 6 already gives uniqueness, but I record the direct argument in the threshold variable of Theorem 1, since both sides of (5)–(6) are decreasing in $\hat{v}$ and single crossing is therefore not obvious. Write $\mathcal{A}(\hat{v}) = m - N \int_0^{\hat{v}} v f$ and $\mathcal{B}(\hat{v}) = \int_{\hat{v}}^{\bar{v}} H(v) f(v) dv$. Differentiating,

$$\mathcal{B}'(\hat{v}) = \hat{v} f(\hat{v}) \ln F(\hat{v}), \quad (\mathcal{A} - \mathcal{B})'(\hat{v}) = -\hat{v} f(\hat{v}) [N + \ln F(\hat{v})],$$

which is positive exactly when $F(\hat{v}) < e^{-N}$. So $\mathcal{A} - \mathcal{B}$ rises and then falls. Since it is positive at $\hat{v} = 0$ and equals $-(N - 1)m < 0$ at $\hat{v} = \bar{v}$, it crosses zero once.

Proof of Proposition 2. With $g_N(K) = m - NK - \mathcal{R}_\mu(K)$, Lemma 6 gives a unique root crossing from above. Since $g_{N+1} < g_N$ pointwise on $(0, m]$, the root falls, $K_{N+1}^* < K_N^*$. As $\mathcal{R}_\mu$ is strictly decreasing on $[0, m)$ and $CRG = \mathcal{R}_\mu(K_N^*)$, the guarantee rises. $\square$

The strict coarseness gap

This subsection proves Proposition 4. Throughout, ψ and K^* are as in (11) and (10), and a *min-R BCCE* is a $\sigma \in \mathcal{BCC}\mathcal{E}(\mu)$ with $R(\sigma) = CRG(\mu)$, which exists by Theorem 1.

Lemma 7 (Rigidity of the minimum). *Every min-R BCCE σ satisfies, σ -almost surely, (i) $a_{(1)} = a_{(2)}$ and (ii) $a_{(1)} = \psi(F(v))$.*

Proof. Both claims concern the joint law of $(v, a_{(1)}, a_{(2)})$, which is invariant under permuting the buyers, and symmetrization preserves revenue and the BCCE property by Lemma 3, so it suffices to prove both claims for exchangeable σ . Exchangeability is what makes $(m - R)/N$ the common rent in part (ii). For (i), if $\Pr_\sigma(a_{(1)} > a_{(2)}) > 0$, Lemma 4 produces a BCCE with strictly lower revenue, contradicting minimality.

For (ii), note that with the top two bids tied, the highest competing bid satisfies $M_i = a_{(1)}$ for *every* buyer i . Buyer i 's deviation payoff therefore depends on the profile only through $(v, a_{(1)})$, and the chain of Step 1 in this appendix applies without change with $a := a_{(1)}$: writing A for the quantile function of $a_{(1)}$ and $K = (m - R)/N$ for the common rent, minimality forces $K = K^*$ and equality throughout (19) and (20). Equality in (20) gives $A(q) = \psi(q)$ for almost every q , so $a_{(1)}$ and $\psi(F(v))$ have the same distribution. Equality in the Hardy–Littlewood step of (19) at almost every

binding level requires $\{a_{(1)} \leq \psi(q)\}$ to coincide, up to null sets, with the lower-contour set $\{v \leq F^{-1}(q)\}$ for almost every $q \in (q_{K^*}, 1]$. To conclude, set $Y := \psi(F(v))$ and pick a countable dense set $Q \subseteq (q_{K^*}, 1]$ of levels at which the displayed coincidences hold. For $q \in Q$ the events $\{a_{(1)} \leq \psi(q)\}$ and $\{Y \leq \psi(q)\}$ agree up to null sets, and since ψ is continuous and strictly increasing on $(q_{K^*}, 1]$, the thresholds $\{\psi(q) : q \in Q\}$ are dense in $(0, \psi(1)]$. Off a single null set, both $a_{(1)}$ and Y are recovered from these events by $a_{(1)} = \inf\{\psi(q) : q \in Q, a_{(1)} \leq \psi(q)\}$ whenever $a_{(1)} > 0$, and likewise for Y , so $a_{(1)} = Y$ on $\{a_{(1)} > 0\} \cup \{Y > 0\}$. For the atom at zero, letting $q \downarrow q_{K^*}$ along Q gives $\{a_{(1)} = 0\} = \{v \leq \hat{v}\} = \{Y = 0\}$ up to null sets. Hence $a_{(1)} = \psi(F(v))$ a.s. $\square$

Lemma 8 (Rigid minima are not interim obedient). *No distribution satisfying (i) and (ii) of Lemma 7 is (the value-action marginal of) a BCE.*

Proof. Recall from Bergemann and Morris (2016) that σ is the value-action marginal of a BCE if and only if, for every buyer i and every measurable deviation map $\xi_i : A_i \rightarrow A_i$, $\mathbb{E}_\sigma[u_i(v, \xi_i(a_i), a_{-i})] \leq \mathbb{E}_\sigma[u_i(v, a)]$. This reduced obedience form is what the argument contradicts. Suppose σ satisfies (i) and (ii). Fix $\eta > 0$ small and consider the region $\mathcal{Z} = [\hat{v} + \eta, \bar{v}]$, on which the margin $m_\eta := \inf_{v \in \mathcal{Z}} (v - \psi(F(v)))$ is strictly positive, since $\psi(F(v)) \leq G(v) < v$ and both functions are continuous. Let $B = \psi(F(\mathcal{Z}))$ be the corresponding set of bid levels, and for $\varepsilon \in (0, m_\eta/2)$ define the deviation map $\xi_\varepsilon(a_i) = a_i + \varepsilon$ if $a_i \in B$ and $\xi_\varepsilon(a_i) = a_i$ otherwise. The map ξ_ε is measurable in the recommendation alone, hence an admissible interim deviation.

Decompose the change in buyer i 's payoff by state. If $a_i \notin B$, nothing changes. If $a_i \in B$ and $a_i = a_{(1)}$, then by (ii) the state is $v = F^{-1}(\psi^{-1}(a_i)) \in \mathcal{Z}$, and by (i) buyer i is tied at the top with some $k \geq 2$ bidders. The deviation converts a $1/k$ share of $v - a_i$ into a win at price $a_i + \varepsilon$, changing the payoff by $(v - a_i)(1 - \frac{1}{k}) - \varepsilon \geq \frac{m_\eta}{2} - \varepsilon > 0$, using $k \geq 2$ and $v - a_i \geq m_\eta$. If $a_i \in B$ but $a_i < a_{(1)}$, the deviation either still loses, changing nothing, or newly wins because $a_{(1)} < a_i + \varepsilon$. In the latter case the state v' satisfies $\psi(F(v')) = a_{(1)} \geq a_i$, so $v' \in \mathcal{Z}$ and the payoff changes by $v' - a_i - \varepsilon \geq v' - \psi(F(v')) - \varepsilon \geq m_\eta - \varepsilon > 0$. Finally, for each ordered pair (i, j) the law of $a_j - a_i$ has at most countably many atoms. Choose ε outside the union of these countable sets, so that $\Pr(a_j = a_i + \varepsilon, a_i \in B) = 0$ for every $j \neq i$ and no new ties arise.

Every state therefore contributes a nonnegative change to the deviating buyer's payoff, and states in which that buyer is tied at the top with $v \in \mathcal{Z}$ contribute at least

$\frac{m_\eta}{2} - \varepsilon$. It remains to find a buyer for whom such states have positive probability, and this does not require symmetry: by (i), in every state with $v \in \mathcal{Z}$ at least two bidders are tied at the top, so $\sum_i \Pr(a_i = a_{(1)}, v \in \mathcal{Z}) \geq 2\Pr(v \in \mathcal{Z})$, and some buyer i^* has $\Pr(a_{i^*} = a_{(1)}, v \in \mathcal{Z}) \geq \frac{2}{N}\Pr(v \in \mathcal{Z}) > 0$. Applying ξ_ε to buyer i^* yields expected gain at least $(\frac{m_\eta}{2} - \varepsilon) \cdot \frac{2}{N}\Pr(v \in \mathcal{Z}) > 0$, so σ is not a BCE. $\square$

Proof of Proposition 4. Suppose $RG_N(\mu) = CRG_N(\mu)$. By hypothesis the interim infimum is attained by some BCE σ_B . Then $R(\sigma_B) = CRG_N(\mu)$, and since $\mathcal{BCE} \subseteq \mathcal{BCCCE}$, σ_B is a min-R BCCE. Lemma 7 gives it properties (i) and (ii), and Lemma 8 then says it is not a BCE, a contradiction. Hence $CRG_N(\mu) < RG_N(\mu)$. $\square$

C Asymptotics

Proof of Theorem 2. From $CRG_N = m - NK_N^* \geq 0$ we get $K_N^* \leq m/N \rightarrow 0$. The integrand in (9) satisfies $0 \leq \frac{(\Phi_\mu(q) - K)^+}{q} \leq \frac{\Phi_\mu(q)}{q} \leq \bar{v}$ and converges pointwise to $\frac{\Phi_\mu(q)}{q}$ as $K \downarrow 0$ whenever $\Phi_\mu(q) > 0$, which holds for every $q > 0$ under Assumption 1. Dominated convergence gives $CRG_N = \mathcal{R}_\mu(K_N^*) \rightarrow \int_0^1 \frac{\Phi_\mu(q)}{q} dq$, and substituting $q = F(v)$ turns this into $\int_0^{\bar{v}} G(v)f(v)dv$, the limiting revenue guarantee of Bergemann, Brooks and Morris (2017). $\square$

Lemma 9 (The interim guarantee converges at rate $1/N$). *For every $N \geq 2$, $0 \leq R_\infty - RG_N \leq \frac{2\bar{v}}{N-1}$.*

Proof. Write $\alpha = \frac{N-1}{N}$. Substituting $t = uF(v)$ in the two conditional means gives

$$G(v) = \int_0^1 F^{-1}(uF(v)) du, \quad \beta_N(v) = \int_0^1 F^{-1}(uF(v)) \alpha u^{\alpha-1} du,$$

so $G(v) - \beta_N(v) = \int_0^1 F^{-1}(uF(v))(1 - \alpha u^{\alpha-1}) du$. The weight $1 - \alpha u^{\alpha-1}$ integrates to zero, is negative for $u < u_0 := (1 - \frac{1}{N})^N$ and positive above, and $F^{-1}(uF(v))$ is nondecreasing in u , so the integral is nonnegative by the Chebyshev integral inequality: $\beta_N \leq G$ pointwise. For the upper bound, $|G - \beta_N| \leq \bar{v} \int_0^1 |1 - \alpha u^{\alpha-1}| du = 2\bar{v}(u_0^\alpha - u_0) = 2\bar{v} \frac{u_0}{N-1} \leq \frac{2\bar{v}}{N-1}$, using $u_0^{\alpha-1} = (1 - \frac{1}{N})^{-1}$. Integrating against f gives the claim. $\square$

Proof of Proposition 3. The bound $0 \leq R_\infty - RG_N \leq 2\bar{v}/(N-1)$ is Lemma 9. The rest of the proof establishes the expansion for $R_\infty - CRG_N$ and then deduces the

one for $RG_N - CRG_N$. Throughout, write $K_N := K_N^*$ for the rent at market size N , $q_N := q_{K_N} = \Phi_\mu^{-1}(K_N)$ for the pooling quantile, and $\hat{v}_N := F^{-1}(q_N)$ for the pooling threshold, so that $\Phi_\mu(q_N) = K_N$ and $q_N = F(\hat{v}_N)$. The proof of Theorem 2 gives $R_\infty = \int_0^1 \frac{\Phi_\mu(q)}{q} dq$ and $K_N \leq m/N \rightarrow 0$. Since Φ_μ is continuous and strictly increasing with $\Phi_\mu(0) = 0$, also $q_N \rightarrow 0$, and since F^{-1} is continuous with $F^{-1}(0) = 0$, also $\hat{v}_N \rightarrow 0$.

Step 1: an exact decomposition. Because Φ_μ is strictly increasing, $\Phi_\mu(q) \geq K_N$ exactly when $q \geq q_N$, so (9) reads $CRG_N = \mathcal{R}_\mu(K_N) = \int_{q_N}^1 \frac{\Phi_\mu(q) - K_N}{q} dq$. Subtracting from R_∞ ,

$$R_\infty - CRG_N = \int_0^{q_N} \frac{\Phi_\mu(q)}{q} dq + K_N \int_{q_N}^1 \frac{dq}{q} = T_N + K_N \ln \frac{1}{q_N},$$

where $T_N := \int_0^{q_N} \Phi_\mu(q)/q dq$. The first term is the value on the pooled quantiles, and the second is the rent on the bidding quantiles. Both are nonnegative.

Step 2: the order of the rent. The fixed-point equation (10) gives $NK_N = m - CRG_N$, and $CRG_N \rightarrow R_\infty$ by Theorem 2, so $NK_N \rightarrow m - R_\infty$. This limit is strictly positive: for $q > 0$, $\Phi_\mu(q)/q$ is the average of the strictly increasing function F^{-1} over $[0, q]$, hence strictly below $F^{-1}(q)$, and integrating over $(0, 1)$ gives $R_\infty < \int_0^1 F^{-1}(q) dq = m$. (This is the sign of $g_N(0)$ in Lemma 6.) Therefore

$$K_N = \frac{m - R_\infty}{N} (1 + o(1)), \quad \text{and in particular} \quad \ln \frac{1}{K_N} = \ln N + O(1). \quad (21)$$

Step 3: the prior near zero. This is the only step that uses the hypothesis on f . Since f is continuous at 0 with $f(0) \in (0, \infty)$ and $F(x) = \int_0^x f(z) dz$, we have $F(x)/x \rightarrow f(0)$ as $x \downarrow 0$. Passing to the inverse, for $t > 0$ put $x = F^{-1}(t)$, so that $t = F(x)$ and $x \downarrow 0$ as $t \downarrow 0$. Then $F^{-1}(t)/t = x/F(x) \rightarrow 1/f(0)$. Fix $\varepsilon \in (0, 1)$ and choose $t_\varepsilon > 0$ such that

$$(1 - \varepsilon) \frac{t}{f(0)} \leq F^{-1}(t) \leq (1 + \varepsilon) \frac{t}{f(0)} \quad \text{for all } t \in (0, t_\varepsilon]. \quad (22)$$

Integrating (22) over $[0, q]$ gives, for all $q \in (0, t_\varepsilon]$,

$$(1 - \varepsilon) \frac{q^2}{2f(0)} \leq \Phi_\mu(q) \leq (1 + \varepsilon) \frac{q^2}{2f(0)}. \quad (23)$$

Step 4: the order of the pooling quantile. Since $q_N \rightarrow 0$, for all N large enough $q_N \leq t_\varepsilon$ and (23) applies at $q = q_N$, where $\Phi_\mu(q_N) = K_N$:

$$(1 - \varepsilon) \frac{q_N^2}{2f(0)} \leq K_N \leq (1 + \varepsilon) \frac{q_N^2}{2f(0)},$$

equivalently $\frac{2f(0)K_N}{1+\varepsilon} \leq q_N^2 \leq \frac{2f(0)K_N}{1-\varepsilon}$. Taking logarithms, $|\ln q_N^2 - \ln(2f(0)K_N)| \leq \ln \frac{1+\varepsilon}{1-\varepsilon}$ for all large N , and since ε is arbitrary,

$$\ln \frac{1}{q_N} = \frac{1}{2} \ln \frac{1}{K_N} - \frac{1}{2} \ln(2f(0)) + o(1) = \frac{1}{2} \ln N + O(1), \quad (24)$$

where the last equality is (21). The square root in $q_N \asymp \sqrt{K_N}$ is the source of the factor $\frac{1}{2}$ in the proposition.

Step 5: the pooled term is of smaller order. For $q \leq q_N$, monotonicity of F^{-1} gives $\Phi_\mu(q) = \int_0^q F^{-1}(t)dt \leq qF^{-1}(q) \leq q\hat{v}_N$, so $0 \leq T_N \leq \hat{v}_N q_N$. For large N , (22) at $t = q_N$ gives $\hat{v}_N = F^{-1}(q_N) \leq (1+\varepsilon)q_N/f(0)$, and Step 4 gives $q_N^2 \leq 2f(0)K_N/(1-\varepsilon)$, so

$$0 \leq T_N \leq \frac{1+\varepsilon}{f(0)} q_N^2 \leq \frac{2(1+\varepsilon)}{1-\varepsilon} K_N = O\left(\frac{1}{N}\right) = o\left(\frac{\ln N}{N}\right).$$

Step 6: the expansion for $R_\infty - CRG_N$. Combining (21) and (24),

$$K_N \ln \frac{1}{q_N} = \frac{m - R_\infty}{N} (1 + o(1)) \left(\frac{1}{2} \ln N + O(1) \right) = \frac{(m - R_\infty) \ln N}{2N} (1 + o(1)),$$

because $O(1)/\ln N \rightarrow 0$. Adding the pooled term from Step 5, which is $o(\ln N/N)$, yields

$$R_\infty - CRG_N = T_N + K_N \ln \frac{1}{q_N} = \frac{(m - R_\infty) \ln N}{2N} (1 + o(1)).$$

Step 7: the cost of coarseness. Write $RG_N - CRG_N = (R_\infty - CRG_N) - (R_\infty - RG_N)$. The first bracket has just been shown to be of exact order $\ln N/N$, while by Lemma 9 the second satisfies $0 \leq R_\infty - RG_N \leq 2\bar{v}/(N-1) = o(\ln N/N)$. Hence $RG_N - CRG_N = \frac{(m-R_\infty)\ln N}{2N}(1+o(1))$ as well, which completes the proof. $\square$

D Reserve Prices

Proof of Proposition 5

Bids in $[0, r)$ never win and never affect the allocation, so normalizing every such bid to 0 changes no payoff and no revenue. Assume this normalization throughout. Symmetrization is unaffected by the reserve, since the allocation rule remains permutation-equivariant, so Lemma 3 applies without change and we may take $\hat{\sigma}$ exchangeable.

Define the Markov kernel Γ_r on $V \times A$ by

$$\Gamma_r((v, a), \cdot) = \begin{cases} \delta_{\{v\}} \otimes \delta_{\{\mathbf{0}\}}, & a_{(1)} < r, \\ \delta_{\{v\}} \otimes \left[\frac{N-1}{N} \delta_{\{\mathbf{a}_{(1)}\}} + \frac{1}{N} \delta_{\{\max\{\mathbf{a}_{(2)}, r\}\}} \right], & a_{(1)} \geq r \text{ and } (a_{(2)} \geq r \text{ or } v \geq a_{(1)}), \\ \delta_{\{v\}} \otimes \left[\frac{N-1}{N} \delta_{\{\mathbf{a}_{(1)}\}} + \frac{1}{N} \delta_{\{\mathbf{0}\}} \right], & a_{(1)} \geq r > a_{(2)} \text{ and } v < a_{(1)}, \end{cases}$$

and let γ be the image of $\hat{\sigma}\Gamma_r$ under one further normalization: on the event where the common bid equals r and $v < r$, replace the common bid by $\mathbf{0}$. The composition is measurable, leaves v unchanged, and places all mass on identical profiles with common bid in $\{0\} \cup [r, \bar{v}]$, so γ is an exchangeable identical play distribution with value marginal μ . By construction, $v \geq r$ holds almost surely on the event that the common bid equals r . Write c for the common bid under γ . Three claims establish the proposition.

Claim 1: revenue weakly falls. On $\{a_{(1)} < r\}$ both revenues are zero. On the second branch, revenue changes from $a_{(1)}$ to $\frac{N-1}{N}a_{(1)} + \frac{1}{N}\max\{a_{(2)}, r\} \leq a_{(1)}$, since $a_{(2)} \leq a_{(1)}$ and $r \leq a_{(1)}$. On the third branch it changes from $a_{(1)}$ to $\frac{N-1}{N}a_{(1)} \leq a_{(1)}$. The final normalization lowers revenue further, from r to 0 on the states it touches.

Claim 2: the buyers' total payoff weakly rises. By (13) the total payoff is $(v - a_{(1)}) \mathbb{1}\{a_{(1)} \geq r\}$. On $\{a_{(1)} < r\}$ it is zero before and after. On the second branch it changes from $v - a_{(1)}$ to $v - \left[\frac{N-1}{N}a_{(1)} + \frac{1}{N}\max\{a_{(2)}, r\} \right] \geq v - a_{(1)}$. On the third branch, where $v < a_{(1)}$ so the sale yields negative surplus, it changes from $v - a_{(1)} < 0$ to $\frac{N-1}{N}(v - a_{(1)})$, which is strictly larger. The final normalization raises the total further on the states it changes, from the negative surplus $v - r < 0$ to 0. Since both distributions are exchangeable, each buyer's payoff is $1/N$ of the total, so $K_\gamma \geq K_{\hat{\sigma}}$.

Claim 3: deviation payoffs weakly fall. A deviation to $\delta < r$ never wins under either distribution and yields zero, while $K_\gamma \geq 0$ because deviating to 0 is available.

So fix $\delta > r$ and let D be the set of such δ that are not atoms of the law of c . The complement is countable and D is dense in $[r, \bar{v}]$. For $\delta > r$, an atom of the averaged law of M_i at δ requires an atom of $a_{(1)}$ or $a_{(2)}$ there, and the kernel maps those to atoms of c at the same point, so levels in D are atom-free for both distributions. For $\delta \in D$ the deviator wins whenever the highest competing bid is strictly below δ , and ties occur with probability zero, so it suffices to compare $\mathbb{E}[(v - \delta)\mathbb{1}\{c < \delta\}]$ with $\mathbb{E}[(v - \delta)\mathbb{1}\{M_i < \delta\}]$ averaged over i .

Couple the two through the randomization in the kernel. Exactly as in Step 1 of Lemma 4, averaging over buyers gives the conditional law of M_i as $\frac{N-1}{N}\delta_{\{a_{(1)}\}} + \frac{1}{N}\delta_{\{a_{(2)}\}}$ on $\{a_{(1)} \geq r\}$, and $M_i \leq a_{(1)} < r$ on $\{a_{(1)} < r\}$. Compare branch by branch.

On $\{a_{(1)} < r\}$, we have $M_i < r \leq \delta$ and $c = 0 < \delta$, so both indicators equal one and the contributions coincide.

On the second branch with $a_{(2)} \geq r$, the kernel gives $c = a_{(1)}$ or $c = a_{(2)}$ with the same weights as M_i , so the two laws coincide and the contributions are equal.

On the second branch with $a_{(2)} < r \leq a_{(1)}$ and $v \geq a_{(1)}$, the coupled pair differs only on the event of probability $\frac{1}{N}$ where $M_i = a_{(2)} < r$ and $c = r$. There $\mathbb{1}\{M_i < \delta\} = \mathbb{1}\{c < \delta\} = 1$ for every $\delta > r$, so the contributions coincide.

On the third branch, $M_i = a_{(2)} < r$ with probability $\frac{1}{N}$ and $c = 0$, and both satisfy the strict inequality against any $\delta \geq r$, so the contributions coincide.

The final normalization moves common bids from r to 0, both strictly below every $\delta \in D$, so it changes no indicator and no contribution above. In every case, then, $\mathbb{E}_\gamma[(v - \delta)\mathbb{1}\{c < \delta\}] \leq \mathbb{E}_{\hat{\sigma}}[(v - \delta)\mathbb{1}\{M_i < \delta\}] \leq K_{\hat{\sigma}} \leq K_\gamma$ for $\delta \in D$.

For an atom of c in $(r, \bar{v}]$, both one-sided limits are available within D and the argument of Lemma 2 applies without change. The deviation to exactly r is the one level not covered by that argument, since there are no admissible deviations below the reserve from which to take the upward limit, and it is checked directly. A deviator bidding r wins when $c = 0$ and ties the full pool when $c = r$, so the deviation payoff is $(1 - \frac{1}{N})A + \frac{1}{N}B$ with $A = \mathbb{E}_\gamma[(v - r)\mathbb{1}\{c = 0\}]$ and $B = \mathbb{E}_\gamma[(v - r)\mathbb{1}\{c \leq r\}]$. Letting $\delta \downarrow r$ through D in the display above gives $K_\gamma \geq B$ by dominated convergence, and the normalization placed $v \geq r$ on $\{c = r\}$, so $B - A = \mathbb{E}_\gamma[(v - r)\mathbb{1}\{c = r\}] \geq 0$ and the deviation payoff is at most $B \leq K_\gamma$. Hence γ is a BCCE. $\square$

The two-pool family is feasible

Lemma 10 (Feasibility). *Fix r , and let (s, q_0) satisfy (14)–(15) with $K = \Phi_\mu(q_0) - rq_0 \geq 0$ and $\Phi_\mu(q_0) - \Phi_\mu(s) \geq r(q_0 - s)$. Then the comonotone identical play distribution with common bid $c = 0$ on the lowest s quantiles of v , $c = r$ on quantiles $[s, q_0)$, and $c = \psi_r(q) := (\Phi_\mu(q) - K)/q$ at quantile $q \geq q_0$ is a BCCE of the auction with reserve r , with revenue R as in (15).*

Proof. The construction has value marginal μ , and by (13) and (15) each buyer's payoff is exactly K . By (14), $\psi_r(q_0) = r$, and the computation in Step 2 of Appendix B shows ψ_r is continuous and strictly increasing on $[q_0, 1]$ when $K > 0$ (when $K = 0$ it is nondecreasing and the argument below is unchanged), so the bid distribution has atoms only at 0 and r . Check every fixed deviation δ .

For $\delta \in (0, r)$ the deviator never clears the reserve and earns $0 \leq K$. For $\delta = 0$ the deviator ties the withholding pool below the reserve, no sale occurs, and the payoff is again $0 \leq K$. For $\delta = r$ the deviator wins the withholding pool and ties the reserve pool, earning

$$\begin{aligned} (\Phi_\mu(s) - rs) + \frac{1}{N}(\Phi_\mu(q_0) - \Phi_\mu(s) - r(q_0 - s)) \\ = K - (1 - \frac{1}{N})(\Phi_\mu(q_0) - \Phi_\mu(s) - r(q_0 - s)) \leq K, \end{aligned}$$

where the identity substitutes (14) and the inequality is the feasibility condition. For $\delta = \psi_r(q)$ with $q \in (q_0, 1]$, the deviator wins exactly the lowest q quantiles at price δ , with no tie mass, earning $\Phi_\mu(q) - \psi_r(q)q = K$: the constraint binds. For $\delta > \psi_r(1) = m - K$ the deviator always wins and earns $m - \delta < K$. Deviations at the atomless levels in between are covered by Lemma 2. Hence the construction is a BCCE, and $CRG_N(r) \leq \widehat{V}_N(r)$, which is part (i) of Theorem 3. $\square$

Comonotonicity is without loss

Lemma 11 (Comonotonization). *For every identical play BCCE of the auction with reserve r there is an identical play BCCE with the same bid marginal, hence the same revenue, in which the common bid is $c = C(F(v))$ for C the quantile function of that marginal.*

Proof. Let (v, c) be the given joint law with bid marginal F_c , and let (v, c°) with $c^\circ = C(F(v))$ be the comonotone coupling of the same marginals. Revenue depends

on the joint law only through F_c and is unchanged. Write $s = \Pr(c = 0)$ and $W = \mathbb{E}[v\mathbb{1}\{c = 0\}]$. Under the comonotone coupling the event $\{c^\circ = 0\}$ coincides up to null sets with the lower set $\{F(v) < s\}$, so $W^\circ = \Phi_\mu(s) \leq W$ by Lemma 5, and the accounting identity (13) gives $K^\circ = (m - W^\circ - R)/N \geq K$ for the payoff.

Every deviation payoff weakly falls. For $\delta \in [0, r)$ it is zero under both couplings. For $\delta \geq r$, the payoff to the fixed bid δ is $(1 - \frac{1}{N})A_\delta + \frac{1}{N}B_\delta$ with $A_\delta = \mathbb{E}[(v - \delta)\mathbb{1}\{c < \delta\}]$ and $B_\delta = \mathbb{E}[(v - \delta)\mathbb{1}\{c \leq \delta\}]$. The probabilities $\Pr(c < \delta)$ and $\Pr(c \leq \delta)$ are fixed by the marginal, and Lemma 5 says the comonotone coupling minimizes both $\mathbb{E}[v\mathbb{1}\{c < \delta\}]$ and $\mathbb{E}[v\mathbb{1}\{c \leq \delta\}]$ among couplings with those probabilities, so $A_\delta^\circ \leq A_\delta$ and $B_\delta^\circ \leq B_\delta$. Every coarse constraint therefore holds at the comonotone coupling, with a weakly larger left side and a weakly smaller right side. $\square$

The chain as an envelope

Lemma 12 (Envelope form). *Let γ be a comonotone identical play BCCE with payoff K , withholding share s , and bid quantile function C . Then*

$$C(q) \geq \underline{C}_K(q) := \max \left\{ r, \frac{\Phi_\mu(q) - K}{q} \right\} \quad \text{for every } q \in (s, 1].$$

Conversely, any comonotone identical play distribution with payoff $K \geq 0$ whose quantile function dominates $\underline{C}_K$ on $(s, 1]$ satisfies $K \geq \mathbb{E}[(v - \delta)\mathbb{1}\{c \leq \delta\}]$ for every $\delta \geq r$.

Proof. Necessity. On $(s, 1]$ the codomain gives $C \geq r$, and $C(q) \geq (\Phi_\mu(q) - K)/q$ is the chain of Step 1 in Appendix B, which applies without change to the deviations $\delta \geq r$ and uses $K \geq 0$, itself the constraint from the deviation to zero. Sufficiency. Fix $\delta \geq r$ and let $x = F_c(\delta)$. Every $q < x$ has $C(q) \leq \delta$, hence $(\Phi_\mu(q) - K)/q \leq \underline{C}_K(q) \leq C(q) \leq \delta$. Letting $q \uparrow x$ and using continuity of Φ_μ gives $\Phi_\mu(x) - \delta x \leq K$, which is the stated inequality. $\square$

Proof of Theorem 3: exactness

For $r \geq m$, the distribution in which every buyer always bids 0 is a BCCE, since a deviation to any $\delta \geq r$ wins and earns $m - \delta \leq 0$, so the guarantee is zero. For $r = 0$ the family constraint (14) reads $K = \Phi_\mu(q_0)$ and, at $s = 0$, (15) is the system (10), so

the claim is Theorem 1 itself. Fix now $r \in (0, m)$. For $s \in [0, 1)$ and $\kappa \in [0, m]$ define

$$h_s(\kappa) := m - \Phi_\mu(s) - N\kappa - \int_s^1 \underline{C}_\kappa(q) dq.$$

Step 1: h_s is concave. The function $q \mapsto \Phi_\mu(q) - rq$ vanishes at 0, decreases on $[0, F(r)]$ and increases thereafter, so its sublevel set at any $\kappa \geq 0$ is an interval $[0, q_r(\kappa)]$ with q_r nondecreasing, and $(\Phi_\mu(q) - \kappa)/q > r$ exactly for $q > q_r(\kappa)$. Hence $h'_s(\kappa) = -N + \ln(1/(s \vee q_r(\kappa)))$, which is nonincreasing in κ .

Step 2: the share bound. Let γ be any comonotone identical play BCCE with share s , payoff K and revenue R . Comonotonicity gives $W = \Phi_\mu(s)$, so (13) reads $R = m - \Phi_\mu(s) - NK$, while Lemma 12 gives $R = \int_s^1 C \geq \int_s^1 \underline{C}_K$. Subtracting, $h_s(K) \geq 0$. By Step 1 the set $\{\kappa \in [0, m] : h_s(\kappa) \geq 0\}$ is an interval, nonempty since it contains K , and since $h_s(m) \leq m - Nm < 0$ its right endpoint $\kappa_+(s)$ is a largest root. So $K \leq \kappa_+(s)$ and

$$R \geq \mathcal{F}(s) := m - \Phi_\mu(s) - N\kappa_+(s).$$

Step 3: admissible shares. The equilibrium's share also satisfies $\Phi_\mu(s) - rs \leq K$. Otherwise, by continuity there is $\eta > 0$ with $(\Phi_\mu(q) - K)/q > r$ on $(s, s + \eta)$, so the envelope forces $C > r$ there, there is no bid atom at r , and $F_c(r) = s$. The chain at $\delta = r$ then reads $K \geq \Phi_\mu(s) - rs$, a contradiction. Call s *admissible* if h_s has a root in $[0, m]$ and $\Phi_\mu(s) - rs \leq \kappa_+(s)$. Every equilibrium share is admissible, so Proposition 5 and Lemma 11 give

$$CRG_N(r) \geq \inf \{ \mathcal{F}(s) : s \text{ admissible} \}.$$

Step 4: candidates and the family. First note that $\kappa_+(s) < m - r$ for every s . At $\kappa = m - r$ the envelope equals r throughout $(s, 1]$, since $(\Phi_\mu(q) - \kappa)/q \leq m - \kappa = r$, so $h_s(m - r) = rs - \Phi_\mu(s) - (N - 1)(m - r)$. The map $s \mapsto rs - \Phi_\mu(s)$ is maximized where $F^{-1}(s) = r$, so $h_s(m - r) \leq rF(r) - \Phi_\mu(F(r)) - (N - 1)(m - r) < 0$ by the hypothesis of the theorem, and since $h'_s(\kappa) = -N$ for $\kappa > m - r$, no root lies at or beyond $m - r$. Hence the crossing $q_r(\kappa_+(s))$ is interior. For admissible s , consider the comonotone distribution with bids 0 on the lowest s quantiles and $\underline{C}_{\kappa_+(s)}$ above. Admissibility says $\kappa_+(s) \geq \Phi_\mu(s) - rs$, equivalently $q_0 := q_r(\kappa_+(s)) \geq s$, so the bids equal r on (s, q_0) and $(\Phi_\mu(q) - \kappa_+(s))/q$ above. By the definition of q_r and continuity,

$\Phi_\mu(q_0) - rq_0 = \kappa_+(s)$, which is (14), while $h_s(\kappa_+(s)) = 0$ is (15). The candidate is therefore a member of the two-pool family with revenue $\mathcal{F}(s)$. Conversely, every family member at parameters (s, q_0) has payoff $K = \Phi_\mu(q_0) - rq_0$, which its feasibility condition makes at least $\Phi_\mu(s) - rs$, and K is a root of h_s by (15), so s is admissible and the member's revenue is at least $\mathcal{F}(s)$. Hence $\widehat{V}_N(r) = \inf\{\mathcal{F}(s) : s \text{ admissible}\}$.

Step 5: attainment and the tie constraint. Lemma 10 shows the candidate at s is a BCCE whenever the average value in the reserve pool is at least r , which holds whenever $F^{-1}(s) \geq r$, that is $s \geq F(r)$. Shares below $F(r)$ are dominated. For $s \leq s' \leq F(r)$,

$$h_{s'}(\kappa) - h_s(\kappa) = \int_s^{s'} (\underline{C}_\kappa(q) - F^{-1}(q)) dq \geq \int_s^{s'} (r - F^{-1}(q)) dq \geq 0,$$

so $\kappa_+(s') \geq \kappa_+(s)$ and $\mathcal{F}(s) - \mathcal{F}(s') = (\Phi_\mu(s') - \Phi_\mu(s)) + N(\kappa_+(s') - \kappa_+(s)) \geq 0$. Moreover, admissibility passes from s to s' , since $\Phi_\mu(s') - rs' \leq \Phi_\mu(s) - rs \leq \kappa_+(s) \leq \kappa_+(s')$. The infimum over admissible shares is therefore unchanged when restricted to $s \geq F(r)$, and there every candidate is a BCCE, so $CRG_N(r) \leq \inf\{\mathcal{F}(s) : s \text{ admissible}\}$. Combining with Steps 3 and 4, $CRG_N(r) = \widehat{V}_N(r)$. $\square$

A lower bound used for continuity

The continuity claim of Theorem 3 remains, since it requires continuity of $\widehat{V}_N$ at $r = 0$. The following relaxation, which is also a closed-form lower bound, gives it.

Lemma 13 (Relaxation). *For $K \in [0, m]$ and $P_0 \in [0, 1]$ define*

$$B_r(K, P_0) := \int_{P_0}^1 \max \left\{ r, \left(\frac{\Phi_\mu(q) - K}{q} \right)^+ \right\} dq,$$

and let

$$L_N(r) := \inf \left\{ \max \{ B_r(K, P_0), m - (N+1)K - rP_0 \} : K \geq (\Phi_\mu(P_0) - rP_0)^+, m - NK \geq B_r(K, P_0) \right\}.$$

Then every identical play BCCE of the auction with reserve $r > 0$ has revenue at least $L_N(r)$.

Proof. Fix an identical play BCCE with common bid c , payoff K , revenue R , and let $s = \Pr(c = 0)$, $P_0 = \Pr(c \leq r)$, $W = \mathbb{E}[v\mathbb{1}\{c = 0\}]$, and C the quantile function of c . Four families of necessary conditions.

Capture, both sides. Taking non-atom levels $\delta \downarrow r$ in (4) and applying dominated convergence, $K \geq \mathbb{E}[(v - r)\mathbb{1}\{c \leq r\}]$. Lemma 5 bounds the right side below by $\Phi_\mu(P_0) - rP_0$, giving the first constraint. Since the atom at r contributes nonnegative value, $\mathbb{E}[v\mathbb{1}\{c \leq r\}] \geq W$, so the same inequality gives $W \leq K + rP_0$.

Chain. For $q \in (s, 1]$ the bid satisfies $C(q) \geq r$, and for $\delta \geq r$ the argument of Step 1 in Appendix B applies without change, giving $C(q) \geq (\Phi_\mu(q) - K)/q$. Hence $C(q) \geq \max\{r, ((\Phi_\mu(q) - K)/q)^+\}$ on $(s, 1]$.

Accounting. By (13), $R = m - W - NK$ with $W \geq 0$.

Combining: $R = \int_s^1 C(q) dq \geq \int_{P_0}^1 \max\{r, ((\Phi_\mu(q) - K)/q)^+\} dq = B_r(K, P_0)$, using $s \leq P_0$ and dropping the nonnegative revenue collected at r on $(s, P_0]$, and $R = m - W - NK \geq m - (N + 1)K - rP_0$ by the bound on W . Feasibility of the equilibrium's own pair (K, P_0) follows from $m - NK \geq m - W - NK = R \geq B_r$. Hence $R \geq \max\{B_r, m - (N + 1)K - rP_0\} \geq L_N(r)$. $\square$

Tightness at zero and continuity

Lemma 14. $L_N(0) = CRG_N(0)$.

Proof. At $r = 0$ the constraint $K \geq \Phi_\mu(P_0)$ is equivalent to $P_0 \leq q_K$, on which the integrand of B_0 vanishes below q_K , so $B_0(K, P_0) = \mathcal{R}_\mu(K)$. Feasibility $m - NK \geq \mathcal{R}_\mu(K)$ is $g_N(K) \geq 0$, equivalently $K \leq K^*$ by Lemma 6. The objective is at least $B_0 = \mathcal{R}_\mu(K) \geq \mathcal{R}_\mu(K^*) = CRG_N(0)$ since $\mathcal{R}_\mu$ is nonincreasing, and the pair (K^*, q_{K^*}) attains it: $\max\{\mathcal{R}_\mu(K^*), m - (N + 1)K^*\} = \max\{CRG_N(0), CRG_N(0) - K^*\} = CRG_N(0)$. $\square$

Proof of Theorem 3, continuity. By Proposition 5 and Lemma 13, $CRG_N(r) \geq L_N(r)$ for every $r > 0$. Also $\widehat{V}_N(0) = CRG_N(0)$ because at $r = 0$ the family constraint (14) reads $K = \Phi_\mu(q_0)$ and (15) reduces, at $s = 0$, to the system (10), whose value is $CRG_N(0)$. Together with Lemma 14 this gives $L_N(0) = \widehat{V}_N(0) = CRG_N(0)$.

Upper semicontinuity at zero. The $s = 0$ members of the family satisfy $K = \Phi_\mu(q_0) - rq_0$ and $m - NK = rq_0 + \int_{q_0}^1 (\Phi_\mu(q) - K)/q dq$. At $r = 0$ this is the system (10), whose solution is unique with a nonvanishing crossing derivative by the computation

following Lemma 6. By the implicit function theorem the solution and its value vary continuously in r near 0, so $\limsup_{r \downarrow 0} CRG_N(r) \leq \limsup_{r \downarrow 0} \widehat{V}_N(r) \leq CRG_N(0)$.

Lower semicontinuity at zero. Let (K, P_0) be feasible for $L_N(r)$. If $m - (N + 1)K - rP_0 \geq CRG_N(0) - r$ the objective already exceeds $CRG_N(0) - r$. Otherwise $K > \frac{m - CRG_N(0)}{N+1} - r \geq K_{N+1}^* - r$, since $m - (N + 1)K_{N+1}^* = CRG_{N+1}(0) \geq CRG_N(0)$, so for r small, $q_K \geq \underline{q} := q_{K_{N+1}^*}/2 > 0$. If $P_0 \leq q_K$ then $B_r \geq B_0 = \mathcal{R}_\mu(K)$ and, as in Lemma 14, feasibility forces $K \leq K^*$ and the objective is at least $CRG_N(0)$. If $P_0 > q_K$, then $\Phi_\mu(P_0) - \Phi_\mu(q_K) = \Phi_\mu(P_0) - K \leq rP_0 \leq r$, and since $\Phi'_\mu \geq F^{-1}(q_K) \geq F^{-1}(\underline{q}) > 0$ on (q_K, P_0) , the interval has length at most $c_0 r$ and the integrand at most $r/\underline{q}$ there, so

$$B_r(K, P_0) \geq \int_{P_0}^1 \left(\frac{\Phi_\mu(q) - K}{q} \right)^+ dq \geq \mathcal{R}_\mu(K) - c_1 r$$

for a constant c_1 depending only on μ and N . Feasibility then gives $g_N(K) \geq -c_1 r$. The crossing slope $g'_N(K^*) = -N + \ln(1/q_{K^*})$ is strictly negative, since a vanishing slope together with concavity would force $g_N \leq 0$ everywhere, contradicting $g_N(0) > 0$. Concavity then gives $K \leq K^* + c_1 r / |g'_N(K^*)|$, and since $|\mathcal{R}'_\mu| \leq \ln(1/\underline{q})$ on the relevant range, the objective is at least $\mathcal{R}_\mu(K) - c_1 r \geq CRG_N(0) - c_2 r$. In all cases $L_N(r) \geq CRG_N(0) - c_2 r$, so $\liminf_{r \downarrow 0} CRG_N(r) \geq \liminf_{r \downarrow 0} L_N(r) \geq CRG_N(0)$. $\square$

E Symmetric Priors

Throughout, π is exchangeable on V^N with per-buyer marginal μ , a BCCE is a joint law σ of $(v_1, \dots, v_N, a_1, \dots, a_N)$ with value marginal π satisfying the coarse constraints, $u_i = q_i(a)(v_i - a_i)$, $M_i := \max_{j \neq i} a_j$, and $K_i := \mathbb{E}_\sigma[u_i]$. Since the good is always allocated, $\sum_i u_i = v_{\text{win}} - a_{(1)}$, where v_{win} is the winner's value.

Two lemmas on the reduction

Lemma 15 (Symmetrization under exchangeable priors). *For a BCCE σ let $\widehat{\sigma} := \frac{1}{N!} \sum_\rho \sigma^\rho$, where σ^ρ is the image of σ under $(v, a) \mapsto (v_\rho, a_\rho)$ with $(v_\rho)_i = v_{\rho(i)}$ and $(a_\rho)_i = a_{\rho(i)}$. Then $\widehat{\sigma}$ is an exchangeable BCCE with $R(\widehat{\sigma}) = R(\sigma)$ and $K_i = K$ for every i .*

Proof. The value marginal of σ^ρ is the image of π under $v \mapsto v_\rho$, which is π by exchangeability. Since $u_i(v_\rho, a_\rho) = u_{\rho(i)}(v, a)$ and $M_i(a_\rho) = M_{\rho(i)}(a)$, buyer i 's equi-

librium and deviation payoffs under σ^ρ are buyer $\rho(i)$'s under σ , so each σ^ρ is a BCCE, and convexity of the constraint set gives the same for the average. Revenue $a_{(1)}$ is permutation invariant, and under $\hat{\sigma}$ every buyer's payoff is the average of the N payoffs under σ . $\square$

Lemma 16 (Pooling under heterogeneous values). *Let $\hat{\sigma}$ be an exchangeable BCCE and let γ be its image under the kernel of Lemma 4, which keeps the value profile and replaces the bids by a common bid c equal to $a_{(1)}$ with probability $\frac{N-1}{N}$ and to $a_{(2)}$ with probability $\frac{1}{N}$. Write $v_{\text{top}} := v_{\text{win}}$, $v_{-\text{top}}$ for the average value of the $N-1$ losers, $\bar{v}$ for the average of all N values, K for the common rent, and $D(\delta)$ for a buyer's payoff from the fixed deviation δ at a non-atom δ of the law of c . Then the identities (16) hold.*

Proof. Under $\hat{\sigma}$ the buyers' total payoff is $v_{\text{top}} - a_{(1)}$, under γ the winner is uniform among the N buyers so it is $\bar{v} - c$, and by exchangeability each buyer's rent is $1/N$ of the total. Since $\mathbb{E}[c] = \frac{N-1}{N}\mathbb{E}[a_{(1)}] + \frac{1}{N}\mathbb{E}[a_{(2)}]$, subtracting gives the first identity. For the second, condition on (v, a) and average over buyers, as in Step 1 of Lemma 4. Under γ every buyer faces $M_i = c$, so the average deviation payoff is $\mathbb{E}[(\bar{v} - \delta)(\frac{N-1}{N}\mathbb{1}\{a_{(1)} < \delta\} + \frac{1}{N}\mathbb{1}\{a_{(2)} < \delta\})]$. Under $\hat{\sigma}$, when $a_{(1)} > a_{(2)}$ the winner faces $a_{(2)}$ and each loser faces $a_{(1)}$, and when the top bids tie every buyer faces $a_{(1)}$, so the average deviation payoff is $\frac{1}{N}\mathbb{E}[(v_{\text{top}} - \delta)\mathbb{1}\{a_{(2)} < \delta\}] + \frac{N-1}{N}\mathbb{E}[(v_{-\text{top}} - \delta)\mathbb{1}\{a_{(1)} < \delta\}]$, where on tie states the two indicators coincide and the expression reduces to the one under γ . Subtracting and using $\bar{v} - v_{\text{top}} = \frac{N-1}{N}(v_{-\text{top}} - v_{\text{top}})$ and $\bar{v} - v_{-\text{top}} = \frac{1}{N}(v_{\text{top}} - v_{-\text{top}})$ gives $\frac{N-1}{N^2}\mathbb{E}[(v_{-\text{top}} - v_{\text{top}})(\mathbb{1}\{a_{(2)} < \delta\} - \mathbb{1}\{a_{(1)} < \delta\})]$, which is the second identity. $\square$

Proof of Proposition 6

Setup. Let γ be an identical play BCCE: an exchangeable joint law of $(v_1, \dots, v_N, c)$ with value marginal π and a single common bid c . Because the uniform tie-break randomizes independently of everything else, buyer i 's equilibrium payoff is $U_i = \frac{1}{N}\mathbb{E}[v_i - c] = \frac{m-R}{N} = K$, exactly as in (3), and coarse obedience for the fixed deviation δ reads, after the limiting argument of (19), $K \geq \mathbb{E}[(v_i - \delta)\mathbb{1}\{c \leq \delta\}]$ for every buyer i .

Lower bound. Lemma 5 applied to the pair $(v_i, \{c \leq \delta\})$ uses only the marginal $v_i \sim \mu$, so $\mathbb{E}[v_i \mathbb{1}\{c \leq \delta\}] \geq \Phi_\mu(\Pr(c \leq \delta))$ regardless of the coupling π . Step 1 in

Appendix B then applies without change, giving $R \geq m - NK^* = CRG(\mu)$, hence $\mathcal{V}_N(\pi) \geq CRG(\mu)$.

Attainment for priors that are common-valued on the bidding region. Suppose $v_1 = \dots = v_N$ π -a.s. on $\{\max_i v_i > \hat{v}\}$, and note that this forces the events $\{v_i \leq \hat{v}\}$ to coincide across buyers, since $v_i > \hat{v}$ implies $v_j = v_i > \hat{v}$. Let the common bid be $c = \psi(F(v_1)) \vee \dots \vee \psi(F(v_N))$, which by the agreement above equals $\psi(F(v_i))$ for every i on the bidding region and 0 on the common pooling event. For any buyer i and any $\delta = \psi(q)$ with $q > q_{K^*}$, the event $\{c < \delta\}$ coincides with $\{v_i < F^{-1}(q)\}$ up to null sets, so the deviation payoff is $\Phi_\mu(q) - \psi(q)q = K^*$ exactly as in Step 2 of Appendix B. The deviation to 0 earns $\frac{1}{N}\mathbb{E}[v_i \mathbb{1}\{c = 0\}] = \frac{1}{N}\mathbb{E}[v_i \mathbb{1}\{v_i \leq \hat{v}\}] = \frac{K^*}{N}$, using the marginal only, and deviations above $\psi(1)$ earn less than K^* as before. Revenue is $\int \psi = CRG(\mu)$, so the bound is attained. Note that the coupling of the values *within* the pooling event is unrestricted: the constraints that would see it never bind.

Necessity. Suppose $\mathcal{V}_N(\pi) = CRG(\mu)$ is attained by some identical play BCCE γ . Minimality forces equality throughout the lower-bound chain: $K = K^*$, the quantile function of c equals ψ almost everywhere, and, for every buyer i , the Hardy–Littlewood step holds with equality at almost every binding level. By Lemma 5, for almost every $q \in (q_{K^*}, 1]$ the event $\{c \leq \psi(q)\}$ coincides up to null sets with $\{v_i \leq F^{-1}(q)\}$, *simultaneously for every i* . Hence for a dense set of thresholds $x \in (\hat{v}, \bar{v}]$ and all i, j , $\{v_i \leq x\} = \{v_j \leq x\}$ up to null sets. Intersecting over rational thresholds, v_i and v_j agree almost surely on the event that either exceeds $\hat{v}$. This is the stated condition.

Finally, failure of attainment implies failure of the infimum itself, by the following stability property of the rearrangement inequality.

Lemma 17 (Stability of the Hardy–Littlewood bound). *Let $v \sim \mu$, fix $q \in (0, 1)$, set $x_q = F^{-1}(q)$ and $L = \{v \leq x_q\}$. If events B_n satisfy $\Pr(B_n) \rightarrow q$ and $\mathbb{E}[v \mathbb{1}_{B_n}] \rightarrow \Phi_\mu(q)$, then $\Pr(B_n \triangle L) \rightarrow 0$.*

Proof. Adding and subtracting x_q ,

$$\mathbb{E}[v \mathbb{1}_{B_n}] - \mathbb{E}[v \mathbb{1}_L] = \mathbb{E}[(v - x_q)(\mathbb{1}_{B_n} - \mathbb{1}_L)] + x_q(\Pr(B_n) - q).$$

On $B_n \setminus L$ we have $v > x_q$ and $\mathbb{1}_{B_n} - \mathbb{1}_L = 1$, and on $L \setminus B_n$ we have $v \leq x_q$ and $\mathbb{1}_{B_n} - \mathbb{1}_L = -1$, so the first expectation equals $\mathbb{E}[|v - x_q| \mathbb{1}_{B_n \triangle L}] \geq 0$, and for any $\gamma > 0$ it is at least $\gamma(\Pr(B_n \triangle L) - \Pr(|v - x_q| < \gamma))$. The left side tends to $\Phi_\mu(q) - \Phi_\mu(q) = 0$

and $x_q(\Pr(B_n) - q) \rightarrow 0$, so $\limsup_n \Pr(B_n \triangle L) \leq \Pr(|v - x_q| < \gamma)$ for every $\gamma > 0$. Letting $\gamma \downarrow 0$ and using continuity of F gives the claim. $\square$

Now suppose identical play BCCEs γ_n had $R(\gamma_n) \rightarrow CRG(\mu)$ for a prior π violating the agreement condition. Then $K_n \rightarrow K^*$, and since the slack in the chain (20) integrates to $R(\gamma_n) - \mathcal{R}_\mu(K_n) \rightarrow 0$, there is a level $q \in (q_{K^*}, 1)$ at which, along a subsequence, the events $B_n = \{c_n \leq \psi(q)\}$ satisfy $\Pr(B_n) \rightarrow q$ and, by the squeeze $\Phi_\mu(\Pr(B_n)) \leq \mathbb{E}[v_i \mathbb{1}_{B_n}] \leq K_n + \psi(q) \Pr(B_n) \rightarrow K^* + \psi(q)q = \Phi_\mu(q)$, also $\mathbb{E}[v_i \mathbb{1}_{B_n}] \rightarrow \Phi_\mu(q)$ for every buyer i . Moreover, q can be chosen at any level where the agreement condition fails. Lemma 17 applied buyer by buyer gives $\Pr(B_n \triangle \{v_i \leq F^{-1}(q)\}) \rightarrow 0$ for every i , forcing $\Pr(\{v_1 \leq F^{-1}(q)\} \triangle \{v_2 \leq F^{-1}(q)\}) = 0$, a contradiction since for the given π this quantity is a fixed positive number at the chosen level. So $\mathcal{V}_N(\pi) > CRG(\mu)$ for such π . $\square$

Proof of Theorem 4

The fixed point. Let $\tilde{g}(K) := m_N - NK - \mathcal{R}_\mu(K) = g_N(K) + (m_N - m)$. By Lemma 6, $\tilde{g}$ is concave on $[0, m]$ with $\tilde{g}(0) \geq g_N(0) > 0$, and $\tilde{g}(m) = m_N - Nm \leq 0$ since $\mathcal{R}_\mu(m) = 0$ and $\max_i v_i \leq \sum_i v_i$. Hence $\tilde{g}$ has a unique zero $\tilde{K} \in (0, m]$, is positive on $[0, \tilde{K})$ and negative on $(\tilde{K}, m]$, and $\tilde{K} \geq K^*$ with strict inequality if and only if $m_N > m$. The following properties of $\tilde{q} = \Phi_\mu^{-1}(\tilde{K})$ will be needed. First, $\mathcal{R}_\mu(\tilde{K}) = \int_{\tilde{q}}^1 \frac{\Phi_\mu(q) - \tilde{K}}{q} dq$, since $\Phi_\mu \geq \tilde{K}$ exactly on $[\tilde{q}, 1]$. Second, because $\Phi_\mu(q)/q$ is the average of F^{-1} on $[0, q]$ and hence at most $F^{-1}(q)$,

$$m_N - N\tilde{K} = \mathcal{R}_\mu(\tilde{K}) \leq \int_{\tilde{q}}^1 F^{-1}(q) dq - \tilde{K} \ln \frac{1}{\tilde{q}} = m - \tilde{K} - \tilde{K} \ln \frac{1}{\tilde{q}},$$

so that

$$\tilde{K} (N - 1 + \ln \tilde{q}) \geq m_N - m \geq 0, \quad (25)$$

with strict inequality on the right when $m_N > m$. In particular $N + \ln \tilde{q} \geq 1$.

The lower bound. Fix a BCCE σ with revenue $R = \mathbb{E}[a_{(1)}]$. For each buyer i , the coarse constraint at a fixed bid δ' that is not an atom of the law of M_i reads $K_i \geq \mathbb{E}[(v_i - \delta') \mathbb{1}\{M_i < \delta'\}]$. Letting $\delta' \downarrow \delta$ through non-atoms and applying dominated convergence and Lemma 5 to the pair $(v_i, \{M_i \leq \delta\})$, which uses only the marginal

$v_i \sim \mu$,

$$K_i \geq \mathbb{E}[(v_i - \delta)\mathbb{1}\{M_i \leq \delta\}] \geq \Phi_\mu(F_{M_i}(\delta)) - \delta F_{M_i}(\delta) \quad \text{for every } \delta \in [0, \bar{v}].$$

This is (19) with M_i in place of c and K_i in place of K , and the argument following it applies without change to the quantile function of M_i , giving

$$\mathbb{E}[M_i] \geq \mathcal{R}_\mu(K_i) \quad \text{for every buyer } i. \quad (26)$$

Since $M_i \leq a_{(1)}$ pointwise, $R \geq \mathbb{E}[M_i]$. Since the winner's value is at most the highest value, $\sum_i K_i = \mathbb{E}[v_{\text{win}} - a_{(1)}] \leq m_N - R$. Let $K_{\min} := \min_i K_i \leq \frac{1}{N} \sum_i K_i$. Then $m_N - NK_{\min} \geq m_N - \sum_i K_i \geq R \geq \mathcal{R}_\mu(K_{\min})$, that is, $\tilde{g}(K_{\min}) \geq 0$, so $K_{\min} \leq \tilde{K}$, and since $\mathcal{R}_\mu$ is nonincreasing, $R \geq \mathcal{R}_\mu(K_{\min}) \geq \mathcal{R}_\mu(\tilde{K}) = m_N - N\tilde{K}$. Taking the infimum over BCCE gives the inequality.

Equality. If $m_N = m$ then $\max_i v_i = v_i$ almost surely for every i , so the prior is common-valued and the claim is Theorem 1. Assume $m_N > m$, so that $N + \ln \tilde{q} > 1$ by (25). Write $P := \{\max_i v_i \leq \tilde{v}\}$ for the pooling event. The agreement hypothesis implies $P = \{v_i \leq \tilde{v}\}$ for every i up to null sets, since $v_i > \tilde{v}$ forces $v_j = v_i > \tilde{v}$ for all j , and on P^c the values are common. Write v for the common value there. The non-degeneracy hypothesis says that with positive probability two buyers have positive values on P , which is equivalent to

$$A := \frac{1}{N} \mathbb{E}[\max_j v_j \mathbb{1}_P] < \mathbb{E}[v_i \mathbb{1}_P] = \Phi_\mu(\tilde{q}) = \tilde{K}, \quad (27)$$

where the middle equality uses $P = \{v_i \leq \tilde{v}\}$ and the first uses $\sum_j v_j \geq \max_j v_j$ on P with strict inequality on a set of positive probability.

For $\varepsilon > 0$ define

$$U_\varepsilon := \tilde{K} - \frac{\varepsilon \tilde{q}}{N + \ln \tilde{q}}, \quad \tilde{c}_\varepsilon(q) := \frac{\Phi_\mu(q) - U_\varepsilon}{q} \quad \text{for } q \in (\tilde{q}, 1],$$

and consider the following exchangeable distribution σ_ε with value marginal π . On P , a buyer whose value is maximal, chosen uniformly at random among the maximizers, bids ε and every other buyer bids 0. Call this buyer *designated*. On P^c every buyer bids $\tilde{c}_\varepsilon(F(v))$. Since $U_\varepsilon < \tilde{K}$, $\tilde{c}_\varepsilon$ is positive on $(\tilde{q}, 1]$ and strictly increasing there, with $\tilde{c}_\varepsilon(\tilde{q}) = (\tilde{K} - U_\varepsilon)/\tilde{q} = \varepsilon/(N + \ln \tilde{q}) =: \eta_\varepsilon \in (0, \varepsilon)$ and $\tilde{c}_\varepsilon(1) = m - U_\varepsilon$.

Revenue and rent. Revenue is $R_\varepsilon = \varepsilon\tilde{q} + \int_{\tilde{q}}^1 \tilde{c}_\varepsilon(q) dq = \varepsilon\tilde{q} + \mathcal{R}_\mu(\tilde{K}) + (\tilde{K} - U_\varepsilon) \ln \frac{1}{\tilde{q}}$. On P the good goes to a maximal-value buyer at price ε and on P^c it is allocated by a tie at price $\tilde{c}_\varepsilon$, so the buyers' total payoff is $\mathbb{E}[\max_j v_j \mathbb{1}_P] - \varepsilon\tilde{q} + \mathbb{E}[(v - \tilde{c}_\varepsilon(F(v))) \mathbb{1}_{P^c}] = m_N - R_\varepsilon$, and by exchangeability each buyer's rent is

$$\frac{m_N - R_\varepsilon}{N} = \frac{m_N - \mathcal{R}_\mu(\tilde{K}) - \varepsilon\tilde{q} + (\tilde{K} - U_\varepsilon) \ln \tilde{q}}{N} = \tilde{K} - \frac{\varepsilon\tilde{q}}{N} \cdot \frac{N + \ln \tilde{q} - \ln \tilde{q}}{N + \ln \tilde{q}} = U_\varepsilon,$$

using $m_N - \mathcal{R}_\mu(\tilde{K}) = N\tilde{K}$. Hence $R_\varepsilon = m_N - NU_\varepsilon = m_N - N\tilde{K} + \frac{N\varepsilon\tilde{q}}{N + \ln \tilde{q}} \rightarrow m_N - N\tilde{K}$ as $\varepsilon \downarrow 0$.

Obedience. Fix a buyer i and a fixed bid δ , and write $D(\delta)$ for the deviation payoff. Under σ_ε , buyer i 's highest competing bid is 0 on P when i is designated, ε on P when i is not, and $\tilde{c}_\varepsilon(F(v)) \geq \eta_\varepsilon$ on P^c . By exchangeability $\mathbb{E}[v_i \mathbb{1}_P \mathbb{1}\{i \text{ designated}\}] = A$ and $\mathbb{E}[v_i \mathbb{1}_P \mathbb{1}\{i \text{ not designated}\}] = \tilde{K} - A > 0$. Choose ε small enough that $U_\varepsilon \geq A$ and $\tilde{K} - A \geq 2\varepsilon\tilde{q}$. Six cases cover $[0, \bar{v}]$.

At $\delta = 0$, buyer i ties only when designated, sharing with $N - 1$ others, so $D(0) \leq A/N \leq U_\varepsilon$.

For $\delta \in (0, \eta_\varepsilon]$, buyer i wins exactly the states of P in which i is designated, so $D(\delta) \leq A \leq U_\varepsilon$.

For $\delta \in (\eta_\varepsilon, \varepsilon)$, write $\delta = \tilde{c}_\varepsilon(q)$ with $q \in (\tilde{q}, 1)$. Buyer i wins the designated states of P and the states of P^c with $\tilde{c}_\varepsilon(F(v)) < \delta$, that is $\tilde{v} < v < F^{-1}(q)$. Adding and subtracting the non-designated states of P , and using $P \cup \{\tilde{v} < v < F^{-1}(q)\} = \{v_i < F^{-1}(q)\}$ up to null sets together with the definition of $\tilde{c}_\varepsilon$,

$$\begin{aligned} D(\delta) &= \mathbb{E}[(v_i - \delta) \mathbb{1}\{v_i < F^{-1}(q)\}] - \mathbb{E}[(v_i - \delta) \mathbb{1}_P \mathbb{1}\{i \text{ not designated}\}] \\ &= U_\varepsilon - \mathbb{E}[(v_i - \delta) \mathbb{1}_P \mathbb{1}\{i \text{ not designated}\}], \end{aligned}$$

so $D(\delta) \leq U_\varepsilon$, since the subtracted term is at least $\tilde{K} - A - \varepsilon\tilde{q} > 0$.

At $\delta = \varepsilon$ the same computation applies, except that in non-designated states of P buyer i now ties with the designated buyer and wins with probability at most $\frac{1}{2}$. The subtracted term becomes at least $\frac{1}{2}(\tilde{K} - A - \varepsilon\tilde{q}) > 0$ and again $D(\varepsilon) \leq U_\varepsilon$.

For $\delta \in (\varepsilon, \tilde{c}_\varepsilon(1)]$, write $\delta = \tilde{c}_\varepsilon(q)$. Buyer i wins every state of P and the states of P^c with $v < F^{-1}(q)$, ties occurring on a null set since $\tilde{c}_\varepsilon$ is strictly increasing and F is continuous. Hence $D(\delta) = \mathbb{E}[(v_i - \delta) \mathbb{1}\{v_i < F^{-1}(q)\}] = \Phi_\mu(q) - \tilde{c}_\varepsilon(q)q = U_\varepsilon$: the constraint binds.

For $\delta > \tilde{c}_\varepsilon(1) = m - U_\varepsilon$, buyer i wins everywhere and $D(\delta) = m - \delta < U_\varepsilon$.

So σ_ε is a BCCE for every small ε , and $CRG(\pi) \leq \inf_\varepsilon R_\varepsilon = m_N - N\tilde{K}$. With the lower bound, this proves the equality claim.

Strict inequality. Suppose BCCE σ_n satisfy $R_n \rightarrow m_N - N\tilde{K} = \mathcal{R}_\mu(\tilde{K})$, and by Lemma 15 take each σ_n exchangeable, so that $K_i^n = K^n$ for all i and the laws of the M_i^n coincide across i . From (26) and $R_n \geq \mathbb{E}[M_i^n]$, $\mathcal{R}_\mu(K^n) \leq R_n \rightarrow \mathcal{R}_\mu(\tilde{K})$, and since $\mathcal{R}_\mu$ is continuous and strictly decreasing on $[0, m]$ by Lemma 6, $\liminf_n K^n \geq \tilde{K}$. From the accounting bound, $K^n \leq (m_N - R_n)/N \rightarrow \tilde{K}$. Hence $K^n \rightarrow \tilde{K}$, and then $\mathbb{E}[M_i^n]$, which is squeezed between $\mathcal{R}_\mu(K^n)$ and R_n , converges to $\mathcal{R}_\mu(\tilde{K})$ as well. From this we get the following. First, $\mathbb{E}[a_{(1)}^n - M_i^n] \rightarrow 0$ with $a_{(1)}^n \geq M_i^n$, so $a_{(1)}^n - M_i^n \rightarrow 0$ in probability. Second, writing A_n for the quantile function of M_i^n , the slack in (26) is $\int_0^1 [A_n(q) - (\Phi_\mu(q) - K^n)^+/q] dq$ with a pointwise nonnegative integrand, and it tends to zero, so along a subsequence $A_n(q) \rightarrow \tilde{\psi}(q) := (\Phi_\mu(q) - \tilde{K})/q$ for almost every q .

Fix such a level $q \in (\tilde{q}, 1)$ and set $B_n := \{M_i^n \leq A_n(q)\}$, with $p_n := \Pr(B_n) \geq q$ by right-continuity. The chain gives $A_n(q) \geq (\Phi_\mu(p_n) - K^n)/p_n \geq (\Phi_\mu(q) - K^n)/q$, and both ends converge to $\tilde{\psi}(q)$. Since $p \mapsto (\Phi_\mu(p) - K)/p$ is strictly increasing for $K > 0$, every limit point of p_n equals q , so $p_n \rightarrow q$. Then $\Phi_\mu(p_n) \leq \mathbb{E}[v_i \mathbb{1}_{B_n}] \leq K^n + A_n(q)p_n \rightarrow \tilde{K} + \tilde{\psi}(q)q = \Phi_\mu(q)$ squeezes $\mathbb{E}[v_i \mathbb{1}_{B_n}] \rightarrow \Phi_\mu(q)$, and Lemma 17 gives $\Pr(B_n \triangle \{v_i \leq F^{-1}(q)\}) \rightarrow 0$ for every i . It remains to replace the buyer-specific events B_n^i by a common one. For $q' < q < q''$ in the almost-everywhere set, the same argument gives $\Pr(M_i^n \leq A_n(q')) \rightarrow q'$ and $\Pr(M_i^n \leq A_n(q'')) \rightarrow q''$, so $\Pr(M_i^n \in (A_n(q'), A_n(q'')]) \rightarrow q'' - q'$. Combined with $a_{(1)}^n - M_i^n \rightarrow 0$ in probability and $A_n(q'') - A_n(q') \rightarrow \tilde{\psi}(q'') - \tilde{\psi}(q') > 0$, this shows that B_n^i and $\{a_{(1)}^n \leq A_n(q)\}$ differ on a set whose probability has lim sup at most $q'' - q'$, which can be made arbitrarily small. Hence $\Pr(B_n^i \triangle B_n^j) \rightarrow 0$ for all i, j , and therefore $\Pr(\{v_i \leq F^{-1}(q)\} \triangle \{v_j \leq F^{-1}(q)\}) = 0$.

If the agreement condition fails, then for some $i \neq j$ the event $\{v_i < v_j, v_j > \tilde{v}\}$ has positive probability, and by Fubini the set of thresholds $x > \tilde{v}$ with $\Pr(v_i \leq x < v_j) > 0$ has positive Lebesgue measure, hence its image under F has positive measure in the quantile variable because F has a positive density, so it meets the almost-everywhere set of levels above. At such a level the displayed probability is positive, a contradiction. This proves the strictness claim. $\square$

Remark 4 (Non-attainment). For $N = 2$ the infimum in the equality case of Theorem 4 is not attained when the values differ with positive probability on P . Attainment

would force equality in $R \geq \mathbb{E}[M_i]$, hence $a_1 = a_2$ almost surely, and equality in $\sum_i K_i \leq m_N - R$, hence a winner with the higher value almost surely. But with tied bids the winner is uniform, so the two values would have to coincide almost surely.

F Atomic Priors, Convex Order, and the Binary Bound

Proof of Theorem 5

For an arbitrary prior, the definitions (8), (9), and (11) are read with the quantile function $F^{-1}(t) = \inf\{x : F(x) \geq t\}$: $\Phi_\mu(q) = \int_0^q F^{-1}(t) dt$ remains convex, nondecreasing, and absolutely continuous, with $\Phi_\mu(0) = 0$ and $\Phi_\mu(1) = m$, and $q_K := \sup\{q : \Phi_\mu(q) \leq K\}$ replaces $\Phi_\mu^{-1}(K)$, with $\Phi_\mu(q_K) = K$ by continuity whenever $q_K < 1$. Proposition 1, Lemma 1, and Lemma 2 nowhere use the density, so it suffices to check the three steps of Appendix B.

The rearrangement inequality. Lemma 5 holds for arbitrary μ with the equality clause weakened: $\mathbb{E}[v\mathbb{1}_B] \geq \Phi_\mu(q)$ for every event with $\Pr(B) = q$, with equality if and only if B agrees almost surely with a lower-contour set $\{v \leq F^{-1}(q)\}$, up to splitting an atom at $F^{-1}(q)$. This is the Hardy–Littlewood inequality in its general form. Step 1 of Appendix B uses only this inequality, the quantile function of the bid, and the monotonicity of $q \mapsto (\Phi_\mu(q) - K)/q$, whose derivative computation requires only that F^{-1} be nondecreasing. The lower bound $R \geq \mathcal{R}_\mu(K)$ therefore holds without change.

Uniqueness. The proof of Lemma 6 never uses the density: the same differentiation gives $\mathcal{R}'_\mu(K) = \ln q_K$ with the sup-definition of q_K , so g_N is concave, positive at 0 by non-degeneracy, negative at m , and its zero $K^* \in (0, m)$ is unique.

Attainment. A deterministic bid $c = \psi(F(v))$ no longer suffices, since at an atom of μ the worst case must randomize the bid. Set instead $v = F^{-1}(U)$ and $c = \psi(U)$ with U uniform on $[0, 1]$. The value marginal is μ , revenue is $\int_0^1 \psi = \mathcal{R}_\mu(K^*) = m - NK^*$, and the rent is K^* . On $(q_{K^*}, 1]$, ψ is absolutely continuous with $\psi'(q) = (F^{-1}(q)q - \Phi_\mu(q) + K^*)/q^2 \geq K^*/q^2 > 0$ almost everywhere, so ψ is continuous, zero on $[0, q_{K^*}]$, and strictly increasing above, and the bid has no atoms except at zero. The three deviation cases of Step 2 are checked with $\{c < \psi(q)\} = \{U < q\}$, so that $\Pr(c < \psi(q)) = q$ and $\mathbb{E}[v\mathbb{1}\{c < \psi(q)\}] = \mathbb{E}[F^{-1}(U)\mathbb{1}\{U < q\}] = \Phi_\mu(q)$ exactly, and the deviation payoff at $\delta = \psi(q)$ is again $\Phi_\mu(q) - \psi(q)q = K^*$. The

deviation to zero ties the atom of mass q_{K^*} and earns $\Phi_\mu(q_{K^*})/N = K^*/N$, and deviations above $\psi(1) = m - K^*$ earn $m - \delta < K^*$. Hence the coupling is a BCCE and $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*) > 0$. $\square$

Proofs for Section 9

Proof of Proposition 9. Suppose $\mu \preceq_{cx} \nu$. By the quantile characterization of the convex order (Shaked and Shanthikumar, 2007, Theorem 3.A.5), the two distributions share the mean m and satisfy $\Phi_\nu(q) \leq \Phi_\mu(q)$ for all q . Hence $\mathcal{R}_\nu \leq \mathcal{R}_\mu$ pointwise by (9), so

$$g_N^\nu(K) = m - NK - \mathcal{R}_\nu(K) \geq m - NK - \mathcal{R}_\mu(K) = g_N^\mu(K) \quad \text{for all } K.$$

By Lemma 6 each is concave with a unique zero, crossing from above, so a pointwise larger function has a weakly larger zero: $K_\nu^* \geq K_\mu^*$. Therefore $CRG(\nu) = m - NK_\nu^* \leq m - NK_\mu^* = CRG(\mu)$.

For the second claim, among distributions supported on $[0, \bar{v}]$ with mean E , the binary distribution $\underline{\mu}$ on $\{0, \bar{v}\}$ is the largest in the convex order: for any such μ and any convex ϕ , $\int \phi d\mu \leq \int \phi d\underline{\mu}$, because on $[0, \bar{v}]$ a convex function lies below the chord joining $(0, \phi(0))$ and $(\bar{v}, \phi(\bar{v}))$. Hence $CRG(\underline{\mu}) \leq CRG(\mu)$ for every such μ . $\square$

Proof of Corollary 1. For $\underline{\mu}$ we have $F^{-1}(t) = 0$ for $t < p_0 := 1 - E/\bar{v}$ and $F^{-1}(t) = \bar{v}$ for $t > p_0$, so $\Phi(q) = \bar{v}(q - p_0)^+$ and $q_K = p_0 + K/\bar{v}$. Then

$$\mathcal{R}(K) = \int_{q_K}^1 \frac{\bar{v}(q - p_0) - K}{q} dq = \bar{v} \int_{q_K}^1 \frac{q - q_K}{q} dq = \bar{v}[(1 - q_K) + q_K \ln q_K].$$

Substituting $K = \bar{v}(q_K - p_0)$ and $E = \bar{v}(1 - p_0)$ into (10) and simplifying yields $(N - 1)(p_0 - q_K) = q_K \ln q_K$, with $CRG = E - N\bar{v}(q_K - p_0)$. Uniqueness of the root is Lemma 6. As $N \rightarrow \infty$ we have $K^* \rightarrow 0$ and $q_K \rightarrow p_0$, so

$$CRG \longrightarrow \bar{v}[(1 - p_0) + p_0 \ln p_0] = E + (\bar{v} - E) \ln \left(1 - \frac{E}{\bar{v}}\right). \quad \square$$

Remark 5. Because the binary prior is atomic, the density-form system (5)–(6) does not apply to it and the worst-case common bid must randomize. This is why Theorem 5 draws the bid through the uniform variable U rather than as a function of the value.

In the limiting case the common bid at $v = \bar{v}$ has CDF $F_c(\delta) = \frac{\bar{v}-E}{E} \cdot \frac{\delta}{\bar{v}-\delta}$ on the support $[0, E]$.